Uncertainty in time-series modeling – An application to the prediction of individual customer lifetime values

Master of Science in Business Analytics

Submitted by
GUTH Hannes

Under the supervision of:
Prof. Dr. Markus Meierer
University of Geneva
Geneva School of Economics and Management

I certify that the work presented here is, to the best of my knowledge and belief, original and the result of my own investigations, except as acknowledged, and has not been submitted, either in part or whole, for a degree at this or any other University.

Signature:

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1 Introduction

For successful businesses, it has always been important to understand their customers, their requirements, behavior and the composition of the whole customer base. Having this understanding provides a competitive advantage when planning marketing and pricing campaigns. To make such initiatives successful and claim necessary resources, it is essential for marketing managers to have an accurate picture of the individual value of each customer, the Customer Lifetime Value (CLV)[1]. Widely used RFM models pose a partial solution to the problem as they order customers according to their predicted individual value but do not provide actual dollar predictions for single customers. More sophisticated tools like the PNBD model (1987) [58] or BG/BB model (2010) [19] fill this gap as they do provide CLV predictions on an individual level. However, also these models are subject to uncertainty that needs to be assessed.

Even though there has been a lot of research about model uncertainty in numerous scientific domains, there has unfortunately been little effort to apply these concepts in marketing models and especially CLV prediction. A closed-form expression of the variance of the CLV, V(CLV), has been derived for the BG/BB model on an individual customer level by McCarthy, Fader, and Hardie in [43]. However, they state that it would be far more complex to derive the same for the more broadly established PNBD model and it remained unclear whether a closed-form solution is at all possible for this model. The R package CLVTools [4] provides a state-of-the-art implementation of the PNBD model that offers to generate prediction intervals (PIs) for individual CLVs, based on a bootstrap procedure.

This work focuses on the PNBD model, the PI-implementation in CLVTools and assesses the derived PIs. In addition, established methods to generate PIs and their variations are implemented to benchmark the bootstrap approach of CLVTools and suggest alternatives. The core objective is to introduce a valid option to correctly assess uncertainty coming from the PNBD model and fill this gap in research. Besides, this work suggests an approach to use the resulting PIs to identify particularly valuable customers.

To achieve these objectives and identify robust and applicable PI-generating methods, a literature review across domains and scientific fields beyond marketing is conducted. The identified methods are deployed on four real-world datasets and their resulting PIs benchmarked against CLVTools' bootstrap implementation using several key metrics. The stability of method performance will be as well subject to scrutiny.

This work is structured as follows. The next section contains a literature overview, covering uncertainty in marketing and giving an overview about existing methods to derive PIs, as well as measures to assess them. The main part will, as a first step, go into detail how the previously introduced methods can be applied in the specific context of the PNBD model and compare them systematically. Besides, also the real-world data sets will be briefly introduced before coming to the central results section. In this core part, methods' performance will be benchmarked and profoundly analyzed. As final step, it will be assessed whether the resulting PIs can be used to help identify particularly valuable customers. The work closes with a brief discussion of results, limitations and areas of possible future research.

[1] In this work, the CLV will not be used directly but a close proxy, or rather indicator, the Conditional Expected Transactions, CET. This step was taken to facilitate the approaches and reasoning at several points without losing significance in results. In addition, one method that is introduced in the course of this work, the Bayesian Approach, is based on an external package that does not include CLV prediction but only CET. It would mean a bigger concession to leave out this method than using CET instead.

2 Literature review

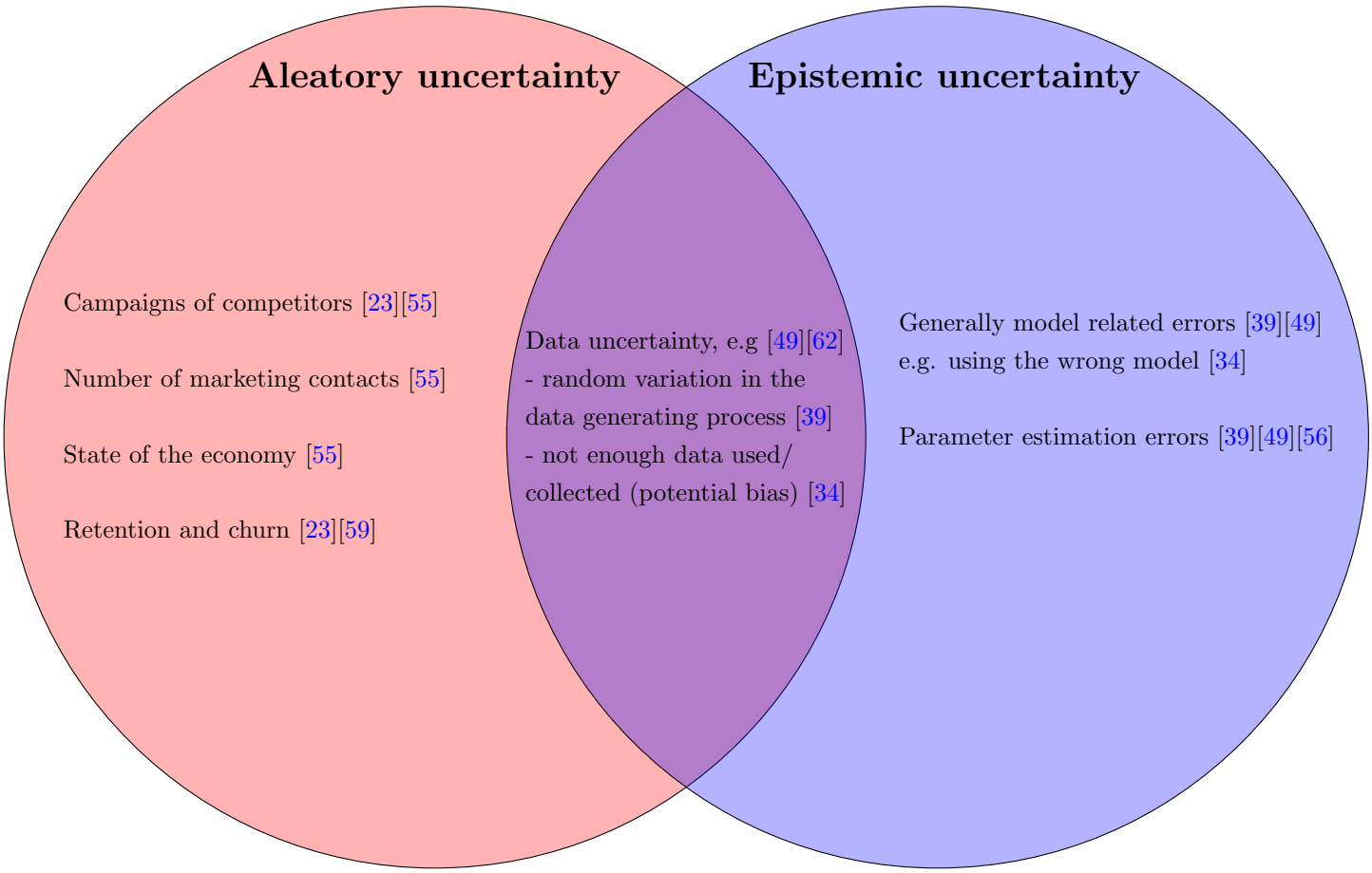
2.1 Review on uncertainty

2.1.1 The role of uncertainty in marketing

To offer a broader, more general view on uncertainty in marketing, the notations of "uncertainty" and "marketing", are clarified first. While no generally recognized definitions exist, there are various attempts to articulate their meaning. Following Collin's dictionary, then "Uncertainty is a state of doubt about the future or about what is the right thing to do." [13] Hubbard states in the context of business "The lack of complete certainty, that is, the existence of more than one possibility. The 'true' outcome/state/result/value is not known." ([30], p. 84) Both definitions agree on the presence of unknown information with respect to a current or potential future state and connected actions (to be taken). On the other hand, there is a similar situation for the definition of marketing. AMA (American Marketing Association) states that: "Marketing is the activity, set of institutions, and processes for creating, communicating, delivering, and exchanging offerings that have value for customers, clients, partners, and society at large." [2], what makes it a very broad field but focused on the placement of offerings to clients or alike. The following paragraph states why uncertainty plays a key role in marketing and why it needs to be considered.
As its name suggests, Marketing is concerned with the placement of offerings on a market and here is where uncertainty comes into play. A market is a place which is heavily concerned with and driven by the actions of its agents, i.e. "customers, clients, partners, and society at large" (AMA, [2]), and their coordination [41]. All of those agents come with uncertainty in their actions as they are ruled by human beings who make (ir-) rational or at least (un-) predictable decisions [20], let it be a new product launch, the choice of a campaign, the location of new branch or simply a consumer's unawareness of a competitor's product which might be superior to their usual choice. Besides those human-driven uncertainties, estimating demand, and placing offerings successfully in the market is affected by additional dimensions of uncertainty, e.g. own product quality [15] that may vary by changing quality of the delivered feedstock. Competitors may bring unexpected technical advancements or the economic situation for the own product can change due to political conflicts and newly imposed taxes, to the good and bad, both ([25], [26]). Also, if a campaign was launched successfully in one region, it does not imply that it will also work out in another one [51]. This list could be extended almost infinitely but should suffice to illustrate why it is important to consider uncertainty in marketing decisions. It is obvious that, from a marketing perspective, it is desirable to keep uncertainty as low as possible to make optimal decisions. Therefore, being able to understand the uncertainty in the specific case, i.e. identify its sources and quantify its amount, is essential. An established approach across science is making predictions about the uncertain future by constructing models which depict a picture of reality, incorporating important aspects and leaving out unimportant ones for simplification. One example is the PNBD model which aims to predict customers' buying behavior and serves as the central model in this work.

2.1.2 Sources of uncertainty

Where does uncertainty arise within CET prediction?
The sources of uncertainty in models in general can be divided into aleatory and epistemic uncertainty [34] what is also valid for the PNBD model. As there is, to my knowledge, no distinct definition of these two concepts, the idea behind shall be briefly explained. Aleatory uncertainty refers to uncertainty coming from random events [40]. It captures e.g. noise in the inherent observations and is therefore input dependent [34]. Uncertainty that comes from within the model, i.e. its parameters, is called epistemic uncertainty [34]. In addition, it "[...] captures our ignorance about which model generated our collected data" ([34], p. 2). Aleatory uncertainty cannot be reduced by collecting more data while this would be possible for epistemic uncertainty [34].



Influences that increase uncertainty of customer behavior, and therefore CET, do so because they happen outside the model and are not (completely) considered in the model, hence they are aleatory. Examples are campaigns of competitors, marketing contacts (in the past, present and future) and state of the economy in a sense that people change their consumption behavior between recession and boom times. Another issue is the possibility of a customer leaving the company forever, e.g. to switch to a competitor or stop consuming. The probability of being "alive" is included in the model but still, most customers will not notify the company when they churn, so it remains a mere probability. The second part addresses epistemic sources. Note that the papers quoted here are not necessarily concerned with CET/CLV prediction but treat forecasting models in general or in a different context, often time series or wind/energy forecasting. Nevertheless, since the PNBD model suffers from similar issues, these aspects are relevant here as well. Particularly often addressed in the literature in association with uncertainty are parameter estimation and data uncertainty, which are both broad fields. The latter is located on the intersection as errors can appear in the data collection and processing (aleatory), and there are potential biases when using the data inside the model (epistemic).

2.1.3 The importance of prediction intervals

With these problems raised, it is evident that mere point forecasts will be in most situations an insufficient indicator for future values as they do not hold information about uncertainty ([11],[46],[31],[22]). Hence, they are often accompanied or even replaced by so-called confidence intervals (for e.g. parameter estimation) and prediction intervals, PIs (in the context of forecasts) [16]. An interval (forecast) is offering a range of possible values of (future) outcomes [36] where the true value will fall into this declared interval with a specified probability. Chatfield ([12]) points out four main points why interval forecasts and PIs are of such importance.

1. They "assess future uncertainty" ([12], p. 476)
2. They enable the user to plan "different strategies for the range of possible outcomes" ([12], p. 476). This means that one can prepare a strategy in case a high value inside the interval is

realized and another one for a low value, or the interval is so narrow and reliable that one can be sure that with e.g. 90% a specific strategy will be appropriate. In the context of CET, it is useful to discover customers with high variability in their CET and therefore target them in particular. There are 2 rationales behind this approach: First, [55] state that there is often a probability distribution for CLVs which has a long right tail.[2] That means that there is upward potential to be realized. Second, [60] state that one should focus on those customers as that offers the opportunity to learn about them and reduce uncertainty.

3. They "compare forecasts from different methods more thoroughly" ([12], p. 476). This means that PIs provide information about the reliability of each method what can be valuable when choosing methods for specific situations.

4. PIs "explore forecasts based on different assumptions more carefully" ([12], p. 476). When there is for example a method that assumes normal distribution, and another method that is similar but does not make this assumption, and they produce different interval lengths, one may want to re-assess the models' assumptions.

Another point, made by [71] (p. 52): "forecasts cannot be expected to be perfect, and intervals emphasize this", underlines perhaps the most important characteristic of prediction intervals, which is reminding forecast users that predictions are probably inaccurate and should be used with appropriate caution. Thinking one step beyond PIs, a more sophisticated option are density predictions, which are comparable with PIs, but assign probabilities to each area inside the interval and hence provide even more information about uncertainty [46].

2.2 Methods to derive prediction intervals

As the importance of PIs has been outlined, this section will focus on their derivation. It is important to note that different models and contexts require different methods to derive PIs. This work focuses on four big classes of methods that are identified by [17]: Bayesian approach, Ensembles, Direct interval estimation and Conformal prediction. In addition, a bootstrap implementation will be used. All methods are introduced in general in this section, explained with the concrete implementation in the CET context in section 3.1 and subsequently benchmarked against each other in section 3.3.

2.2.1 Bootstrap method

The Bootstrap method is the method which is used as a benchmark. It is a non-parametric and powerful approach to estimate statistics like the mean or quantiles of a distribution and therefore as well PIs. The general approach to conduct a bootstrap goes as follows.

Step 1: From a sample of data of size n, draw n times an entry with replacement to create a new sample of size n

Step 2: Repeat Step 1 sufficiently often, e.g. 1000 times to create 1000 new samples

Step 3: For each of these new 1000 samples, calculate the desired metric.

Step 4: From this distribution of the metric, take the central x% of predictions, resulting in the desired interval.

The central assumptions for bootstrapping are that the initially sampled data, from which the new samples are created, are representative for the whole population and i.i.d.

2.2.2 Mini Bootstrap / Ensemble

Ensembles are in general a very straightforward method to derive PIs. They can be described as follows: "An ensemble is a collection of a (finite) number of neural networks or other types of predictors that are trained for the same task. A combination of many different predictors can often improve predictions [...]" ([61], p.190). When one has enough fitted models and therefore enough point predictions,

[2] And therefore, CET as main driver for CLV can be expected to be also right-tailed.

one can 1. construct naïve prediction intervals [27] or 2. calculate mean and variance, assume a (normal) distribution and derive PIs by calculating the respective z-values for the desired quantile [17]. Svetunkov et al. suggest a special form of this approach, which has characteristics of both, Ensembles and Bootstrap [64]:

Step 1: Fit 1 single model that has several parameters

Step 2: Derive the covariance matrix of the parameters of this fitted model

Step 3: Derive a large number, e.g. 100, of parameter combinations that, together, have the characteristics described in the covariance matrix (one has to make assumptions about their distribution)

Step 4: Treat these parameter combinations as independent models and make predictions with these models for each record of the data set

Step 5: Take the naïve prediction intervals for each record

This approach has similarity with the previously described bootstrap approach and therefore, similar performance is to expect. In contrast, only 1 model is fitted and from there, all other models are derived by simply simulating parameters what makes it computationally more attractive.

2.2.3 Bayesian Approach

The roots of Bayesian statistics go back to the 18th century, to Thomas Bayes, as the name suggests [76]. It is hard to tell who was the first to make use of this approach to construct PIs but one of the pioneers in this field was John Aitchison in 1964 [1] who introduced the idea of using the Bayesian approach's strength in forming tolerance regions. The idea is to derive a probability distribution over the parameter(s) and based on this, derive a distribution of the outcome and take the desired statistics from that outcome. The process is described in the following, based on [1], [17] and [33].

Step 1: It is useful to have information about the parameter probability distribution (prior distribution) before the parameters are observed (or use an uninformative distribution) [1]

Step 2: It is necessary to have a likelihood function that describes how likely it is to observe the data that are revealed step by step under the current parameter distribution [17]

Step 3: As more information (data) is revealed, update the prior parameter distribution with the new information to eventually derive the posterior parameter distribution, using Bayes' rule [17]

Step 4: Predict the outcome based on the posterior parameter distribution to get a distribution of the outcome (posterior predictive distribution)

Step 5: Calculate the PIs based on this outcome distribution [17]

The Bayesian Approach is a method to estimate parameters and at the same time delivers a distribution of outcomes from where one can derive the PIs. Both, the posterior parameter distribution and the posterior predictive distribution, can be retrieved approximately by applying the Markov Chain Monte Carlo Method (MCMC).

2.2.4 Quantile regression

Quantile regression was first introduced by Koenker and Basset in 1978 [37] and is a form of direct interval estimation. Therefore, it does not model a distribution of outcomes but is designed to directly output an interval [17].
Following the process described in [17], [37] and [38], Quantile regression works through optimizing the parameters of a model with respect to a loss function that is employed during the model fitting process. The idea is that there is one combination of parameters that yields a specific number of overpredictions and underpredictions of the true outcomes. Aiming e.g. for a balanced result would mean 50% overpredictions and 50% underpredictions (assuming a continuous distribution where point

predictions never hit their target perfectly). [38] state that this idea applies for any quantile other than the 50% quantile as well: A loss function penalizes deviation from the desired above-below ratio, which is 1:1 in the 50% case. Targeting intervals, e.g. a 90% interval, would require finding the 5%- and 95% quantile. Therefore, two parameter combinations must be found. Applying the same principle as above, the loss function would penalize according to the desired quantiles, yielding the parameter combinations that come closest to the objective, i.e. 5% and 95% over- and underpredictions respectively.[3]
This approach requires interrupting and changing the model fitting procedure. As that would be out of scope for this work, a modified version will be implemented for the CET context that keeps the core idea but simplifies the procedure drastically. A step-by-step guide for the concrete implementation will be provided in section 3.1.4.

2.2.5 Conformal prediction

Conformal prediction, or Conformal Inference, is a relatively young method to derive prediction intervals with attractive empirical guarantees and few assumptions about the data and the model form to which it is applied [3]. It was first introduced by Vovk, Gammermann and Saunders in [57] and [67] and raised a lot of attention in recent years. Conformal prediction has two main forms of implementation, Full conformal Prediction (Transductive Conformal Prediction) and Split Conformal Prediction (Inductive Conformal Prediction). Full CP has been developed first and Split CP has emerged as an important special case [3] after being initially introduced in [48] in 2002. The importance of the split version comes from the high computational costs associated with the full version but also sacrifices statistical efficiency [3]. In this work, the focus will be put exclusively on the split version, because of the mentioned computational efficiency but also, what is crucial, because Full CP is not applicable for the PNBD model. The reason for this will be outlined in section 3.1.5.

Split conformal prediction
Even though CP is applicable to both, regression and classification problems, the focus of this work is by default only on regression.
General procedure for split conformal prediction in regression, following [3], [48], [65]:

Step 1: Split the data in training, calibration and test set

Step 2: Fit the prediction model on the training set

Step 3: Make predictions with this model on the calibration set

Step 4: "Identify a heuristic notion of uncertainty [...]" ([3], p. 5), e.g. the absolute error of a prediction $|y_i - \hat{f}(x_i)|$

Step 5: Define a score function (A score function can be chosen arbitrarily as long as it has the right orientation, i.e. lower absolute values are better [65]) and apply this function to the forecasting errors of the calibration set

Step 6: Take the desired (1-α)-quantile $\hat{q}$ of the scaled errors as defined in [3], p. 4

$$\hat{q} = \lceil (n+1)(1-\alpha) \rceil / n \tag{1}$$

Step 7: Make predictions on the test set

Step 8: Use the previously calculated quantile to form prediction intervals (add/subtract the quantile to/from the point predictions made on the test set)

Regardless of the score function, these intervals have the coverage guarantee, defined in [67] (concrete formulation taken from [3], p.6)

$$P\left(Y_{test} \in \boldsymbol{C}(X_{test})\right) \geq 1 - \alpha \tag{2}$$

[3] Note that a combination that yields 5% underpredictions automatically yields 95% overpredictions and would therefore be a good candidate for predicting the upper boundary values.

The only condition that must hold for this coverage guarantee is exchangeability [3] in a sense that records, from training, validation and test are exchangeable, which is weaker than i.i.d. as exchangeability can be expressed as follows (see [65], p.3.)

$$(Y_1, ..., Y_{n+1}) \stackrel{d}{=} \left(Y_{\sigma(1)}, ..., Y_{\sigma(n+1)}\right), \quad for\ all\ permutations\ \sigma \tag{3}$$

It implies that the distribution after splitting the data randomly is expected to be the same in each split.
The concrete implementation of CP for the PNBD model will be shown in the section 3.1.5.

2.3 Measures to assess prediction intervals

One core objective of this work is to assess the performance of prediction intervals derived from different methods and benchmark them against each other. To achieve this goal, several measures will be introduced in the following. They mainly address reliability and sharpness and combined performance. Table 1 informs about recent works which employed these measures and how they are useful.

List of abbreviations

n	Number of customers
y_i	True value
PI_i	Prediction interval for customer i
UB_i	Upper interval boundary for customer i
LB_i	Lower interval boundary for customer i
$pred_i$	Prediction for customer i

PICP (Prediction Interval Coverage Probability[4])

This measure holds the percentage of cases when the true value falls inside the constructed PI.

$$PICP = \frac{1}{n} * \sum_{i=1}^{n} \mathbb{1}(y_i \in PI_i) \tag{4}$$

ACE (Average Coverage Error[5])

This measure indicates how much on average the Prediction Interval Nominal Confidence, PINC, i.e. 90%, and the actual coverage, PICP, differ. Lower values are preferable.

$$ACE = PINC - PICP \tag{5}$$

PICPW (Prediction Interval Coverage Probability Weighted)

PICPW weighs the coverage of customers with their true number of transactions and, therefore, overweighs important customers compared to the neutral PICP. For example, if there are 1,000 repeat purchases across all customers in a dataset, a customer with 15 purchases, whose interval covers these 15, would have this "1" for "True" or "value covered" weighted by 15/1,000. In contrast, if a customer made 0 transactions, it would not increase PICPW, regardless of whether the interval covers this value, because its weight is 0/1,000. Hence, PICPW measures whether high-value customers are identified, placing less importance on low-value customers. Another, interpretation of this measure is that it essentially represents the coverage of repurchases across the customer base. Higher values are preferable.

[4]Different names have been found in the literature for this measure: [50]: Coverage; [28]: True coverage; [29], [70], [35]: PICP; [42]: Coverage rate; [69]: ECP (Empirical coverage probability)
[5]Different names have been found in the literature for this measure: [29], [70], [69]: ACE; [42]: ACD (absolute coverage difference)

$$PICPW = \sum_{i=1}^{n} \mathbb{1}(y_i \in PI_i) * \frac{y_i}{\sum_{j=1}^{n} y_j} \tag{6}$$

PIARW (Prediction Interval Average Relative Width)

This measure assesses the interval width with respect to the level of the prediction. It takes into account the special CET situation where the predictions have significantly different values and hence, intervals must be assessed accordingly. E.g. an interval having the width of 4 is of different value for a prediction of 1 and a prediction of 50. The PIARW is the mean over all customers. Lower values are preferable.

$$PIARW = \frac{1}{n} * \sum_{i=1}^{n} \frac{UB_i - LB_i}{pred_i} \tag{7}$$

PIARWW (Prediction Interval Average Relative Width Weighted)

Like PICPW, PIARWW puts higher value to more important customers, i.e. to their relative interval widths. In contrast, all PIARWs of customers with > 0 repurchases are considered, not just those which actually cover the true value. Lower values are preferable.

$$PIARWW = \sum_{i=1}^{n} \frac{UB_i - LB_i}{pred_i} * \frac{y_i}{\sum_{j=1}^{n} y_j} \tag{8}$$

MSIS (Mean Scaled Interval Score)

MSIS is based on the interval score proposed in [21], but scaled by the prediction and averaged over all customers. In other time-series contexts in the literature, see table 1, the scaling is done with the seasonal differences which is not applicable here. This measure assesses reliability and sharpness at the same time because it penalizes interval width and true values outside of PIs. Lower values are preferable.

$$MSIS = \frac{1}{n} * \sum_{i=1}^{n} \frac{UL_i - LB_i}{pred_i} + \frac{2}{\alpha} * \left(\mathbb{1}(y_i > UB_i) * \frac{y_i - UL_i}{pred_i} + \mathbb{1}(y_i < LB_i) * \frac{LB_i - y_i}{pred_i} \right) \tag{9}$$

SWR (Sharpness Width Ratio)

This measure evaluates coverage per width achieved by the intervals. Higher values are preferable.

$$SWR = \frac{PICP}{PIARW} \tag{10}$$

Upper coverage

This measure indicates the percentage of times the upper boundary was not exceeded by the true value.

$$Upper\ coverage = \frac{1}{n} * \sum_{i=1}^{n} \mathbb{1}(y_i \leq UB_i) \tag{11}$$

Lower coverage

This measure indicates the percentage of times the lower boundary was lower than the true value.

$$Lower\ coverage = \frac{1}{n} * \sum_{i=1}^{n} \mathbb{1}(y_i \geq LB_i) \tag{12}$$

Computational time

This measure holds the time taken by a method to calculate the PIs on a dataset. The time must be interpreted with caution because methods can be implemented with varying effort and precision, e.g. the bootstrap method can have 100 or 1000 bootstrap samples what changes the time a lot. More information on this will be provided in section 3.3 of this work.

Summary

The following table is not exhaustive in a sense that it does not contain all works that dealt with evaluating their PIs but contains a collection of recent studies that employed inter alia measures that are applicable for the CET context as well (left). On the right part, the targeted characteristics of each measure are indicated.

	[50]	[28]	[29]	[70]	[42]	[35]	[21]	[75]	[69]	[76]	[44]	**This work**	Reliability and Sharpness	Reliability	Sharpness	Downside risk	Upside potential	Context independent Generalizability
PICP	x	x	x	x	x	x		x	x	x		x		x				x
ACE			x	x	x				x			x		x				x
PICPW												x		x			x	x
PIARW								x		x		x			x			
PIARWW												x			x			
MSIS	x				x		x				x	x	x					
SWR												x	x					x
Upper coverage	x											x		x			x	x
Lower coverage												x		x		x		x
Comp. time												x						x

Table 1: Overview about measures to assess PIs

3 Main part

3.1 Application of PI-generating methods

In this section, it will be illustrated how the previously introduced methods are implemented in the PNBD context. The used data sets will be introduced in section 3.2.
A few notes to make before going into detail with the single methods:

- For Bootstrap, Mini bootstrap / Ensemble and the Bayesian method, only customer data from a current customer cohort are used without explicitly mentioning it.
- For Quantile regression and Conformal prediction, in addition to the current cohort[6], an old cohort is necessary. It will be clearly indicated at any point which cohort is meant.
- Each cohort has a holdout and a prediction period that is the same across methods but differs across data sets. Find more information on the cohorts in tables 3 to 6.
- This work uses $\alpha = 0.1$, therefore 90% prediction intervals are targeted. Any other quantile would be possible as well.

3.1.1 Bootstrap (BS)

The bootstrap approach is used as main benchmark for all other methods, as it is the most established method, spread in all scientific domains and straight-forward to implement and understand. The procedure is implemented in the R package CLVTools [4] and works as follows.

Step 1: From a customer base of length n, sample n times a customer ID with their respective transactions and save the draw in a new sample. (sampling with replacement)

Step 2: Repeat Step 1 m times to receive m bootstrap samples

Step 3: For each bootstrap sample, fit a new PNBD model on the training period (leaving all settings constant, e.g. start parameters)

Step 4: With each model, predict the CET for all customers for the holdout period

Step 5: Take the $\frac{\alpha}{2}$ and $1 - \frac{\alpha}{2}$ quantiles of these predictions to receive the PI boundaries on an individual customer level

For more detailed information regarding this procedure, refer to CLVTools [4].

3.1.2 Mini bootstrap / Ensemble (EN)

As the name suggests, this method is a combination of two methods. Its idea has been initially introduced by Svetunkov et al. in [64] and the implementation in the PNBD context follows their approach.

Step 1: Fit one single PNBD model on the training period of all customers

Step 2: Receive parameter estimates and their respective covariance matrix

Step 3: With the information about the estimates and covariance matrix, simulate n (i.e. 100) draws of parameters, assuming the parameters to be multivariate normal distributed, so that the totality of simulations mimics the initial estimates and the covariance matrix.

Step 4: Each draw of this parameter simulation is the basis for a new model

Step 5: Run the prediction for the holdout period with each of these new models

[6] In practice, the current cohort is the cohort of customers which joined 1 or 2 years ago and for whom the future behavior is predicted. In this work, also the current cohorts lay in the past for all data sets, so that it can be assessed whether the methods construct PIs that do indeed cover the true observations.

Step 6: Receive n values for the CET

Step 7: Take the $\frac{\alpha}{2}$ and $1-\frac{\alpha}{2}$ quantiles of these CET predictions to receive the PI boundaries on an individual customer level

Its advantage compared to the pure bootstrap is that it requires only a single model fit and is hence less computationally intense.

3.1.3 Bayesian method (BA)

This approach requires to fit the PNBD model with the Bayesian approach and then take the intervals from the posterior predictive distribution. As re-estimating the model with the Bayesian approach would be out of scope for this work, the existing implementation within the R package BTYDplus [52] is used. Therefore, only a rough explanation how the Bayesian model fitting works will be given. For more detailed information, refer to the mentioned package.

Step 1: Estimate the PNBD model with the Bayesian approach: (BTYDplus package)

- No previous information about the parameter distribution is given
- Use the MCMC method to get the posterior parameter distribution
- From this distribution, use again the MCMC method to get the draws of the posterior predictive CET distribution

Step 2: Take the $\frac{\alpha}{2}$ and $1-\frac{\alpha}{2}$ quantiles of the posterior predictive CET distribution to receive the PI boundaries on an individual customer level

3.1.4 Quantile regression (QR)

As indicated before, the implementation of Quantile regression will be modified from the "original" approach but keeps the idea of direct interval estimation. To avoid introducing an optimization within the model fitting process, the optimization is 1. conducted after the model fitting and 2. broken down into to a grid search. In addition, for this method to work, it is necessary to have an old and a current cohort, each with a training and holdout period. The reason for this issue will be clarified in the following procedure and the explanation below.

This first part is conducted on an old cohort.

Step 1: Build a grid of potential parameter combinations *1:M*

Step 2: Each parameter combination is the basis for a new model m

Step 3: Predict the CET for each customer with every new model m

Step 4: Each model will create a certain amount of overpredictions and underpredictions, while it is the objective to find those models that yield $\frac{\alpha}{2}*100\%$ and $(1-\frac{\alpha}{2})*100\%$ overpredictions. To assess how good each model performs, introduce an absolute distance measure that serves as a loss function for the

- Upper boundary parameters (only $\frac{\alpha}{2}$ of the true values should be *above* the predictions)

$$loss_upper_m = \left| \left(\frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(y_i > pred_{i,m}) \right) - \frac{\alpha}{2} \right| \tag{13}$$

- Lower boundary parameters (only $\frac{\alpha}{2}$ of the true values should be *below* the predictions)

$$loss_lower_m = \left| \left(\frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(y_i < pred_{i,m}) \right) - \frac{\alpha}{2} \right|. \tag{14}$$

The first part of the equations measures exactly the coverage and then $\frac{\alpha}{2}$ is deducted.

Step 5: Collect the calculated differences

Step 6: Select the parameter combinations m_u, m_l that yield the lowest absolute differences, one combination for the upper, one combination for the lower boundary.

It is apparent that the true values y_i are required in order to calculate the coverage each model yields. In reality, one does either not have these values (because they lay in the future, and they shall be predicted) or they are known because they lay in the past. Then, there would be no point in predicting them. To overcome this issue, the previous procedure was carried out on an old cohort and now one assumes that customer behavior for one firm or data set will not change a lot over time. In this case, there is no reason to assume that the optimal parameter combinations, which yield the desired quantiles, would change. In summary, the first part is done on old customers of the company to figure out the optimal parameter combination for each boundary and then those combinations are used on the new, interesting data to construct PIs.

This second part is hence conducted on the new cohort.

Step 1: Fit a PNBD model on the training period

Step 2: Make predictions for a holdout (unknown) period

Step 3: Make also predictions on the holdout period using the two previously derived models m_u, m_l

Step 4: The resulting predictions from Step 3 are the upper and lower interval boundaries

A few notes on modifications departing from the general description for the first part of the procedure:

- For setting up the grid as narrow as possible and by this minimizing computational effort, it is helpful to have some prior knowledge where parameters should be located approximately. This step was done manually, running several attempts by hand for a rough orientation and then providing alternative values in this area. It will turn out that, regardless of the dataset (and in section 3.3.3, regardless of the learning and holdout period lengths), similar parameter combinations are selected. This is a convenient situation for the application in practice as the combinations do not need to be identified over and over again, what makes up most of the computational costs, or the approach can be run on a smaller grid to get more accurate values.
- Many customers have no repurchases after their initial purchase. No matter, which parameter combination is selected, the model will never predict exactly 0 (or negative) repurchases, what causes an issue for the lower boundary, which can only come very close to 0 but will never include 0. It makes sense to introduce a small tolerance and set those predictions to 0 which are reasonably close to 0 to give the method a chance to perform well. This seems arbitrary but in practice, one could argue that a managerial decision regarding a customer will barely differ if CET = 0 or CET = 0.1 (what is the used tolerance). Also, one could claim that it would be "unfair" to the other methods. Quantile regression is the only method that suffers from this problem and adding this tolerance increases at the same time the QR-intervals' width, so it comes at a cost. Doing this trade-off for other methods would not increase their performance.

Find a visualized description for the application of QR here.

3.1.5 Conformal prediction (CP)

As indicated above, only Split Conformal prediction will be implemented because the full version is computationally much more intense and not applicable for the PNBD model. The following paragraph outlines why this is the case by the help of a basic example:

Full conformal prediction
Following [3] and [65], full CP is implemented as follows: Assume there are 250 records of a) predictors X_i = 1:250 and b) observed outcomes Y_i = 1:250. From these 250 records, 1 complete record is taken out. Assuming to not know what the true Y for this record is, one can only state that this outcome

lives in the range of all possible future outcomes $\boldsymbol{Y}$. The approach is to take n values as possible outcomes out of $\boldsymbol{Y}$ and reunite each of these n "invented" records with the 249 unchanged ones, ending up again with n sets of records. For each set, a new model is fitted what is computationally costly. When predicting with each of these different models the value that was left out, and applying a score function to this outcome, one ends up with n score values. From here, one would go on and create prediction intervals. However, in the context of the PNBD model, it is not possible to continue because it does not consider the true outcomes when fitting the model, as it is exclusively focused on the purchase history of customers. Therefore, fitting n models by supplying n different outcomes for Y would not lead to different models and would in turn not allow to form PIs.

The implementation of Split CP follows in principle the steps from the general description but there are several modifications to be made.

1. Heteroskedasticity of the outcomes: It appears that customers have a very different repurchasing behavior and might buy again 0 or 50 times within the same dataset. When the model is off by 3, say for the first customer ($y_1 = 0$), it predicts 3 and for the second customer ($y_2 = 50$) it predicts 53, the absolute delta will be equal, but the model would have done a bad job for the first customer and good job for the second customer. Assuming that the absolute error is increasing with the CET level, it is reasonable to employ PIs that are adaptive. Otherwise, the method would suffer from overcoverage for small and under-coverage for large CET values. A possible solution is given in [3] and [65], as they suggest to scale the residuals by e.g. their standard deviation, "studentization". For reasons of simplicity, the used scale will be the expected absolute difference between true value and prediction (prediction error), regressed exclusively on the level of the prediction.[7] The process of deriving the model that predicts the error is conducted on the old cohort and is straight forward:

   Step 1: Fit the PNBD model on the training data

   Step 2: Make predictions on the holdout period

   Step 3: Get the absolute differences for each prediction and its true value

   Step 4: Fit a linear model: $absolute_difference_i \sim CET_i$

   For every CET, there is now a reasonable scale available.

2. As it was the problem with Quantile regression before, Conformal prediction needs the true data as well, not only for the scaling but the actual method functionality, too. Again, one can make the assumption that customers' behavior for a firm is approximately constant over time. The whole process of model fitting, and derivation of the quantile and scaling is therefore conducted on the old cohort where the true transactions are known. The quantile and the scaling model are then forwarded to the current cohort.

3. In contrast to the standard procedure in 2.2.5, it is not necessary to use a validation set to get the quantiles. In the standard procedure, a model is fitted on training data and the quantile is derived from a validation set. If the quantile were taken from predictions on the known training data, it would be too small because the model knows these data, leading to underperformance on unknown data. This is not the case in the PNBD model because, in order to predict the CET of the new cohort, a new model is fitted on exactly those new customers. Therefore, it is expected that the quantiles fitted on the "known" training customers will be adequate for customers which are "known" to the new model. In other words, the model used to get the quantile knows its customers and the model on which the quantiles are applied, knows its customers as well, hence the quantiles should be adequate from this point of view.

[7] This is a strong assumption but reasonable to some extent, see figure 11a - 11d, especially in absence of a valid alternative.

Assuming that the linear model for the standard deviation has been derived already, the whole process is summarized in the following:

For the old cohort

Step 1: Split the data set into a training and a test set (split customer wise)

Step 2: Fit the PNBD model on the training set

Step 3: Make point predictions on this training set and the test set

Step 4: Take the absolute differences between the predictions from the training set and the true outcomes

Step 5: Scale the differences (divide by the customers' respective estimated absolute difference from the previously fitted linear model)

Step 6: Take the desired (1-α)-quantile of the scaled residuals with equation 1 as defined in [3], p. 4 $\hat{q} = \lceil (n+1)(1-\alpha) \rceil / n$

Step 7: Rescale the quantile by multiplying with the individual linearly estimated absolute difference for each customer i in the test set: $quantile * estimated_absolute_difference(CET_i)$

Step 8: Add/subtract these individually scaled quantiles to/from the point predictions on the test set to get the PIs

Step 9: Evaluate the coverage on the test set

Theoretically, it is not necessary to split the old data set and conduct the test from step 5 to step 7, but it is reasonable to check if the quantile works at least with old data before transferring it to new data.

For the new cohort

Step 1: Train the PNBD model on the new training data

Step 2: Make point predictions for the holdout (unknown) period

Step 3: Rescale the quantile from the old cohort with the estimated absolute difference for each customer i: $quantile * estimated_absolute_difference(CET_i)$

Step 4: Add/subtract these individually scaled quantiles to/from the point predictions on the new data to get the PIs

Conformal prediction Repeated (CR)
To overcome a potential bias when splitting into training and test set, i.e. when the Exchangeability assumption is violated, a second version of CP is implemented in this work. It will be called Conformal prediction Repeated, CR. It is basically the same procedure but the splitting into training and testing and the connected quantile generation is repeated n times. Subsequently, the average over all n quantiles is taken to avoid the mentioned potential bias. It is only a side implementation and will not be discussed any further but might become relevant for applications with a very limited database.

Important remark for Quantile regression and Conformal prediction: When assuming temporal consistency of parameters (QR) and quantiles (CP), it is valid for both directions. This means that for long time customers, for whom there is no previous cohort, the parameters/quantile taken from a newer cohort should be valid as well when estimating their CET uncertainty. One might justifiably argue that their behavior cannot be expected to be the same as for new customers because we know that they 1. have been loyal for a long time already and 2. might indeed have a higher likelihood to physically die when considering they have been customers in e.g. a pharmacy for 40 years.

Find a visualized description for the application of CP and CR here.

3.1.6 Conceptual comparison of the methods

Table 2 gives an overview about the general characteristics and assumptions of the single introduced approaches.

Method		Focused uncertainty	Central assumptions made	True values needed	Approach complexity	Frequentist approach
Bootstrap		Epistemic	In this context none	No	Medium	Yes
Ensemble		Epistemic	Normal distribution of re-sampled parameters	No	Low	Yes
Bayesian		Epistemic, Aleatory	None but knowledge about prior parameter distribution can enhance results	No	High	No
Quantile regression		Epistemic, Aleatory	Constant customer behavior over time	Yes	Medium	Yes
Conformal prediction	CP	Epistemic, Aleatory	• Exchangeability • Constant customer behavior over time	Yes	Medium	Yes
	CR	Epistemic, Aleatory	Constant customer behavior over time	Yes	Medium	Yes

Table 2: Conceptual method comparison

3.2 Data

The introduced methods will be deployed on four real-world data sets, containing transactions from customers of different retailers. As QR, CP and CR require past data with training and prediction period, an old cohort and a new cohort will be created for all data sets, each with training and prediction periods that are equal for all methods. The learning periods for all data sets and cohorts are approximately 1 year. The holdout period for the gift and electronics data set are also approximately 1 year for each cohort and 2 years for the multichannel and the apparel retailer. For more details regarding the data sets, see the tables below. These tables follow mainly the data presentations in [5].

Gift retailer ("gift")

	Old learning	Old holdout	Total old	New learning	New holdout	Total new
Customers	-	-	2124	-	-	2064
Transactions	6103	2229	8332	6721	2408	9129
Available timeframe and split in weeks	52	52	104	52	52	104
Average number of purchases per customer	2.87	1.05	3.92	3.26	1.17	4.42
Standard deviation of repeated purchases	4.71	5.91	7.34	4.13	7.16	6.92
Zero repeaters	956	221	750	867	199	700
First entry date [8]	-	-	08.12.2002	-	-	08.12.2004
Last entry date	-	-	15.12.2002	-	-	15.12.2004

Table 3: Gift data set

[8] First and Last entry date define the timeframe how the respective cohort was formed, i.e. for the gift data set, the old cohort is comprised of customers who made their first purchase between 08.12.2002 and 15.12.2002.

Electronics retailer ("el")

Metric	Old learning	Old holdout	Total old	New learning	New holdout	Total new
Customers	-	-	728	-	-	4859
Transactions	3206	1037	4243	20679	5707	26386
Available timeframe and split in weeks	52	52	104	52	52	104
Average number of purchases per customer	4.4	1.42	5.83	4.26	1.17	5.43
Standard deviation of repeated purchases	4.18	5.06	5.92	4.44	5.73	6.1
Zero repeaters	183	58	138	1264	208	1058
First entry date	-	-	01.01.2000	-	-	01.01.2002
Last entry date	-	-	31.03.2000	-	-	30.03.2002

Table 4: Electronics data set

Multichannel retailer ("multi")

Metric	Old learning	Old holdout	Total old	New learning	New holdout	Total new
Customers	-	-	3644	-	-	3885
Transactions	7365	1198	8563	7632	720	8352
Available timeframe and split in weeks	52	104	156	60	104	164
Average number of purchases per customer	2.02	0.33	2.35	1.96	0.19	2.15
Standard deviation of repeated purchases	1.92	2.6	2.44	1.97	1.75	2.23
Zero repeaters	2020	212	1823	2223	138	2103
First entry date	-	-	01.12.2005	-	-	01.12.2008
Last entry date	-	-	31.12.2005	-	-	31.12.2008

Table 5: Multichannel data set

Apparel retailer ("apparel")

	Old learning	Old holdout	Total old	New learning	New holdout	Total new
Customers	-	-	814	-	-	2836
Transactions	1725	1395	3120	5561	4923	10484
Available timeframe and split in weeks	52	104	156	52	104	156
Average number of purchases per customer	2.12	1.71	3.83	1.96	1.74	3.7
Standard deviation of repeated purchases	1.62	2.75	3.76	1.58	2.82	3.66
Zero repeaters	383	150	256	1545	544	904
First entry date	-	-	01.01.2000	-	-	01.01.2003
Last entry date	-	-	15.01.2000	-	-	15.01.2003

Table 6: Apparel data set

3.3 Results

3.3.1 General method performance

The performance of all methods across datasets and metrics is summarized in table 7 below and analyzed in the following sections.

Method	Data	PICP	ACE	PICPW	PIARW	PIARWW	MSIS	SWR	Upper coverage	Lower coverage	Time in sec (abs)	Time (rel)
BS	gift	0.0165	0.8835	0.0440	0.2704	0.2896	37.4091	0.0609	0.7602	0.2563	267.6303	65.1644
	el	0.0025	0.8975	0.0098	0.2193	0.1828	93.1505	0.0113	0.8137	0.1887	207.8255	50.6027
	multi	0.0028	0.8972	0.0250	0.3912	0.3701	32.6382	0.0072	0.9223	0.0806	792.8348	193.0446
	apparel	0.0190	0.881	0.0315	0.1357	0.1103	21.6792	0.1403	0.6393	0.3798	157.2298	38.2833
EN	gift	0.0107	0.8893	0.0285	0.2516	0.2614	36.8537	0.0424	0.7602	0.2505	61.9928	15.0944
	el	0.0033	0.8967	0.0117	0.1670	0.1454	93.6913	0.0197	0.8142	0.1891	94.3064	22.9623
	multi	0.0003	0.8997	0.0023	0.2433	0.2055	33.7451	0.0011	0.9197	0.0806	148.2455	36.0958
	apparel	0.0201	0.8799	0.0325	0.1380	0.1116	21.6375	0.1457	0.6396	0.3805	56.5614	13.7719
BA	gift	0.9840	-0.0840	0.8176	11.4655	9.5828	13.6366	0.0858	0.9840	1.0000	881.9382	214.7401
	el	0.8843	0.0157	0.4911	4.5810	5.1226	76.0336	0.193	0.8845	0.9998	105.1194	25.5952
	multi	0.9768	-0.0768	0.6023	9.4248	8.6909	16.747	0.1036	0.9768	1.0000	7994.939	1946.6601
	apparel	0.963	-0.0630	0.8509	4.8186	4.0753	6.8631	0.1998	0.9672	0.9958	133.2052	32.4337
QR	gift	0.9520	-0.0520	0.6003	8.0923	6.5505	12.1046	0.1176	0.9520	1.0000	274.6686	66.8781
	el	0.9337	-0.0337	0.4799	26.6989	19.5407	44.726	0.0350	0.9341	0.9996	253.7952	61.7957
	multi	0.9843	-0.0843	0.6455	18.214	14.3973	21.1006	0.054	0.9843	1.0000	268.4811	65.3715
	apparel	0.9076	-0.0076	0.6088	3.7018	2.7961	6.781	0.2452	0.9087	0.9989	218.2475	53.1403
CP	gift	0.9549	-0.0549	0.6171	6.0977	5.2581	11.7084	0.1566	0.9549	1.0000	7.3405	1.7873
	el	0.8891	0.0109	0.4247	13.8910	10.6413	49.8796	0.0640	0.8891	1.0000	11.8448	2.8840
	multi	0.9853	-0.0853	0.6795	11.6406	9.9927	16.5662	0.0846	0.9853	1.0000	30.8859	7.5203
	apparel	0.8801	0.0199	0.6338	2.8058	2.4097	7.565	0.3137	0.8946	0.9855	4.1070	1.0000
CR	gift	0.9549	-0.0549	0.6171	6.0977	5.2581	11.7084	0.1566	0.9549	1.0000	72.9781	17.7692
	el	0.8891	0.0109	0.4247	13.8910	10.6413	49.8796	0.0640	0.8891	1.0000	140.1040	34.1134
	multi	0.9853	-0.0853	0.6795	11.6406	9.9927	16.5662	0.0846	0.9853	1.0000	243.5311	59.2966
	apparel	0.8801	0.0199	0.6338	2.8058	2.4097	7.5650	0.3137	0.8946	0.9855	34.7923	8.4715

Table 7: Overall method performance

The results from table 7 are summarized by method, averaging the respective results across datasets (table 8) and by a ranking of these averages (table 9).

Method	PICP	ACE	PICPW	PIARW	PIARWW	MSIS	SWR	Upper cov.	Lower cov.	Time [s] (abs)
BS	0.0102	0.8898	0.0276	0.2542	0.2382	46.2193	0.0549	0.7839	0.2264	356.3801
EN	0.0086	0.8914	0.0188	0.2000	0.1810	46.4819	0.0522	0.7834	0.2252	90.2765
BA	0.9520	-0.0520	0.6905	7.5725	6.8679	28.3201	0.1456	0.9531	0.9989	2278.8005
QR	0.9444	-0.0444	0.5836	14.1768	10.8212	21.1781	0.1130	0.9448	0.9996	253.7981
CP	0.9274	-0.0274	0.5888	8.6088	7.0755	21.4298	0.1547	0.9310	0.9964	13.5446
CR	0.9274	-0.0274	0.5888	8.6088	7.0755	21.4298	0.1547	0.9310	0.9964	122.8514

Table 8: Average method performance

Method	PICP	ACE	PICPW	PIARW	PIARWW	MSIS	SWR	Upper cov.	Lower cov.	Time [s] (abs)
BS	5	5	5	2	2	5	5	5	5	5
EN	6	6	6	1	1	6	6	6	6	2
BA	1	4	1	3	3	4	3	1	2	6
QR	2	3	4	6	6	1	4	2	1	4
CP	3	1	2	4	4	2	1	3	3	1
CR	3	1	2	4	4	2	1	3	3	3

Table 9: Methods ranking by metric (smaller ranks are better)

Major findings

1. The two bootstrap-based methods (BS, EN) deliver consistently undercoverage across data sets while the four other methods (BA, QR, CP, CR) deliver roughly the desired coverage.
2. The widths of the two underperforming methods are significantly lower than the widths of the other four methods.
3. There is no trade-off method that combines both strengths or finds a compromise.
4. CP is by far the fastest method, followed by EN and CR. All other methods are significantly slower, BA is the most time consuming.[9]

Additional findings

1. Overweighting high value customers (PICPW) decreases the performance of the four methods with good overall coverage (BA, QR, CP, CR) significantly in absolute and relative numbers.[10] For BS and EN, it enhances their performance but their coverage remains very low.

[9] Regarding the interpretation of computing times, it should be noted that times vary srongly on how many e.g. bootstrap samples (BS), parameter simulations (EN), parameter combinations (QR) or repetitions (CR) are used, what can change computation time tremendously. In addition, QR is the only method has been implemented with parallel computing what decreases its computation time. For companies who once figured out the perfect quantile (CP) or perfect parameter combinations (QR) for their customers, CP and QR will be a lot faster because only their respective second part has to be done. Hence the times rather give a rough idea about computational intensity for all methods with a reasonable number of runs/samples/parameter combinations and no prior knowledge.

[10] Roughly between 58% and 70% of overall transactions have been covered, to employ the alternative interpretation of PICPW.

2. EN and BS have the by far lowest width with respect to the prediction level (PIARW), followed by BA, CR and CP, serving a middle way. QR has in two data sets comparable values with these three methods but in the electronics and the multichannel case, it produces very wide intervals, resulting in a high average. It seems that the performance of QR has a higher dependency on the data set.

3. PIARW shrinks in tendency for all methods for more valuable customers. This observation might be partially driven by the lower likelihood of customers having a high number of actual transactions when having a very small CET at the same time. (compared to customers with a lower CET which are more likely to have a lower number of actual transactions) Scaling the interval width with a CET < 1 will inflate it, causing higher PIARW for customers with lower CET. This effect may dominate the fact that the PIs of lower CET customers are typically shorter than the ones of higher CET customers, especially thinking of CETs <0.1.

4. The combined assessment of sharpness and width in MSIS sees BS and EN on the lower rank, due to the penalization of non-coverages. The rest of the methods have roughly the same performance, with a slight disadvantage for BA.

5. The same picture can be seen in SWR. The 4 reliable methods have around 2-3x more coverage per width than BS and EN. Here, it is QR that has a slight disadvantage.

6. CP and CR hit the desired 90% the most accurate. (least over- or undercoverage)

7. The reliable methods, except BA, produce wider intervals for the electronics data set while not achieving a higher coverage at the same time, what leads to the conclusion that their performance is to some extent data set dependent.

8. BS and EN have a very high variability (coefficient of variation) in reliability measures. For sharpness measures, all methods have comparable variability, see variation table (table 15 in the appendix which contains the coefficient of variation across datasets for each method and measure.

Figure 1 supports the above findings. One can clearly observe the two different types of methods, the bootstrap-based ones with low coverage and short intervals on the left and the other methods on the right, delivering the desired coverage at the cost of wider intervals. Across methods, there is no consensus which data set or its respective results contain most uncertainty, i.e. which need the widest intervals. However, every method delivers approximately consistent PICP across data sets.

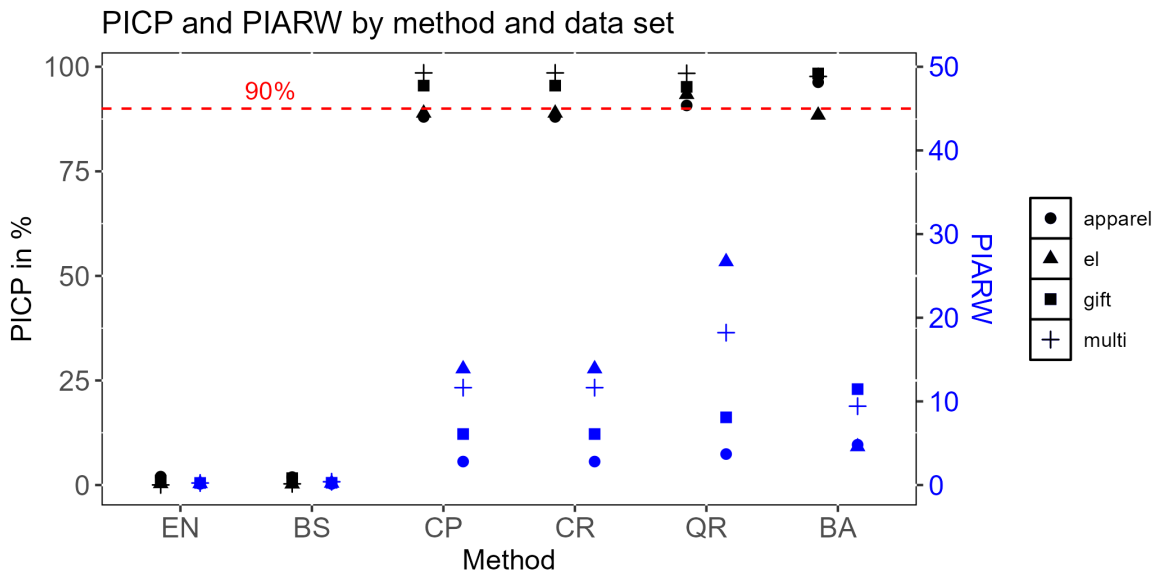


Figure 1: PICP and PIARW by method and data set

Figure 2 shows the concrete lengths of prediction intervals and the true number of transactions exemplary for selected customers from the apparel data set. Again, one can observe the small ranges that are covered by BS and EN and the wide spread of true observations that show how unlikely it is for these methods to cover a true value. The other methods capture the uncertainty appropriately and, at least in this selection, cover all true points. For this data set and selected range of CET, BA has the longest intervals, but that is not representative for the rest of the data sets and other ranges of CET as the overview table 7 shows. In this plot, only customers with a very similar CET were selected to ensure a reasonable scaling on the y-axis. This mentioned, except BA, all methods produce across customers very similar intervals that appear to depend exclusively on CET, what is valid for all data sets, see also figures 10a - 10c. This finding raises concerns about how individually the PIs are constructed for each customer. More on this topic in chapter 3.4.

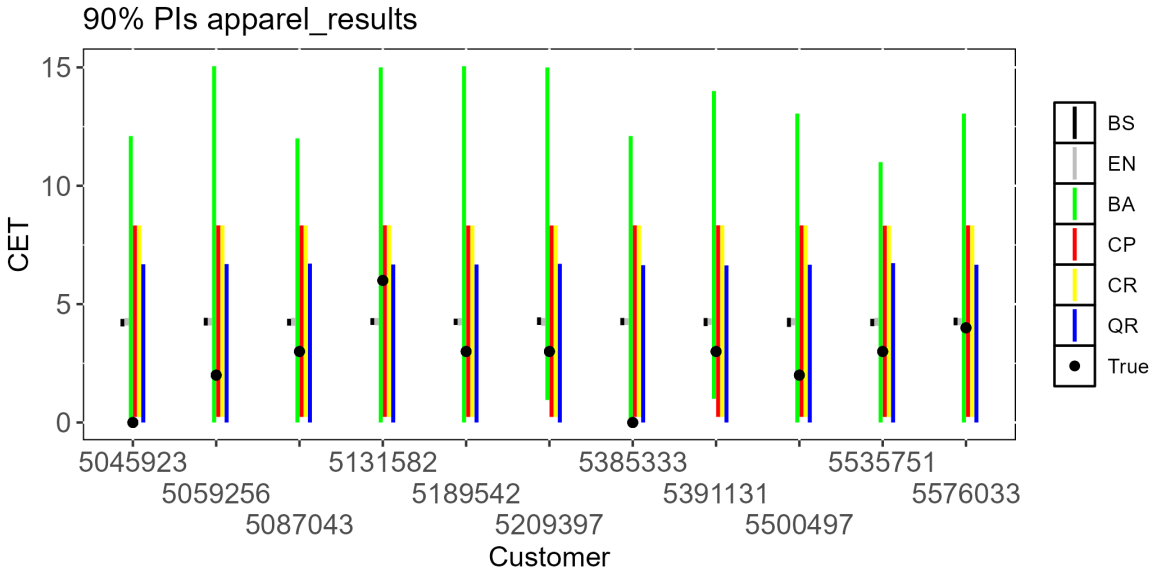


Figure 2: 90% PIs apparel results

Summary
There are two types of methods, those with wide intervals and appropriate coverage and those with narrow intervals and distinct undercoverage. In combined measures of reliability and sharpness, usually the methods with high coverage outperform the other two. Find a metric-wise summary visualized in figure 3 below.

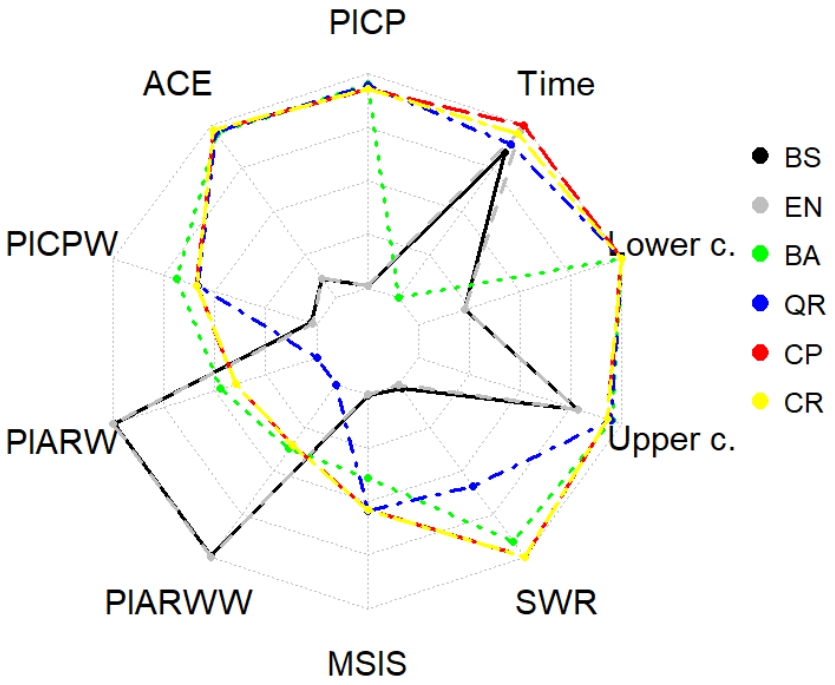
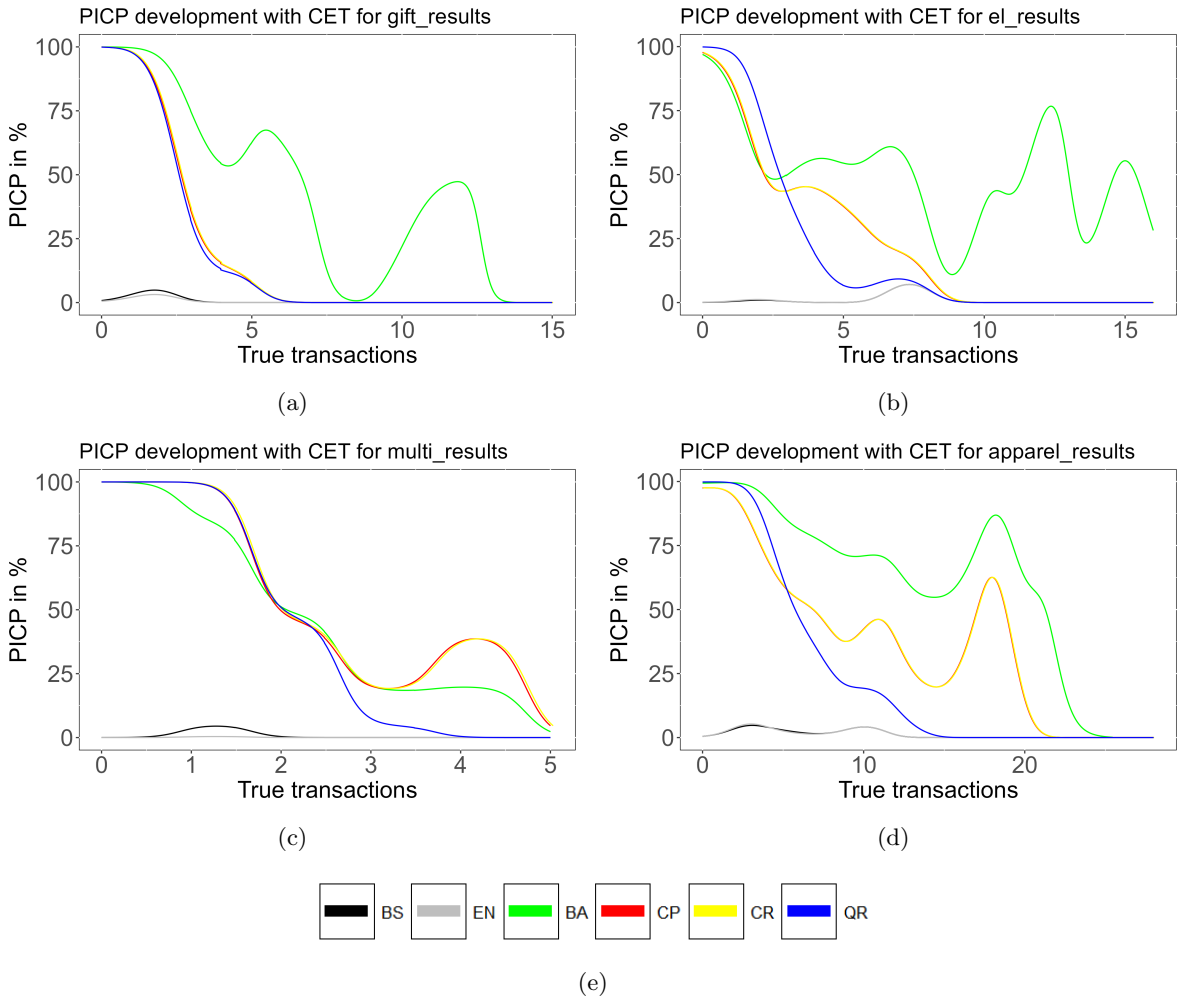


Figure 3: Method characteristics

3.3.2 Coverage and width across CET levels

In the previous section, it was mentioned that methods might perform differently at different levels of CET. This will be examined more in-depths.

Coverage
Figures 4a - 4d smooth the PICP development across CET by the help of a kernel with normal distribution. This development is not quite the same for different data sets.
However, across all data sets, BA has the highest coverage what is not surprising as is has the highest overcoverage. BS and EN have the lowest coverage, close to 0, at all levels, see figures 12a - 12d in the appendix which show the development especially for BS and EN with an adapted scale. Another note to make is generally a downward tendency that is approximately true for all methods on all data sets. This means that the intervals are less likely to include the true value if it is high. After the analysis of PICPW, this comes with no surprise but poses an issue as it is exactly this segment that managers are interested in. The high coverage rate at lower values comes from many true values being 0 and usually intervals are covering 0 either by nature or by the small tolerance that was given in case of QR. The lower coverage on high values is visible across methods what leads to the conclusion that the model might be less accurate in this area. Indeed, the relative model error $\left(\frac{|pred_i - y_i|}{pred_i}\right)$ for higher values is by tendency higher than for smaller values (see figure 11 in the appendix).



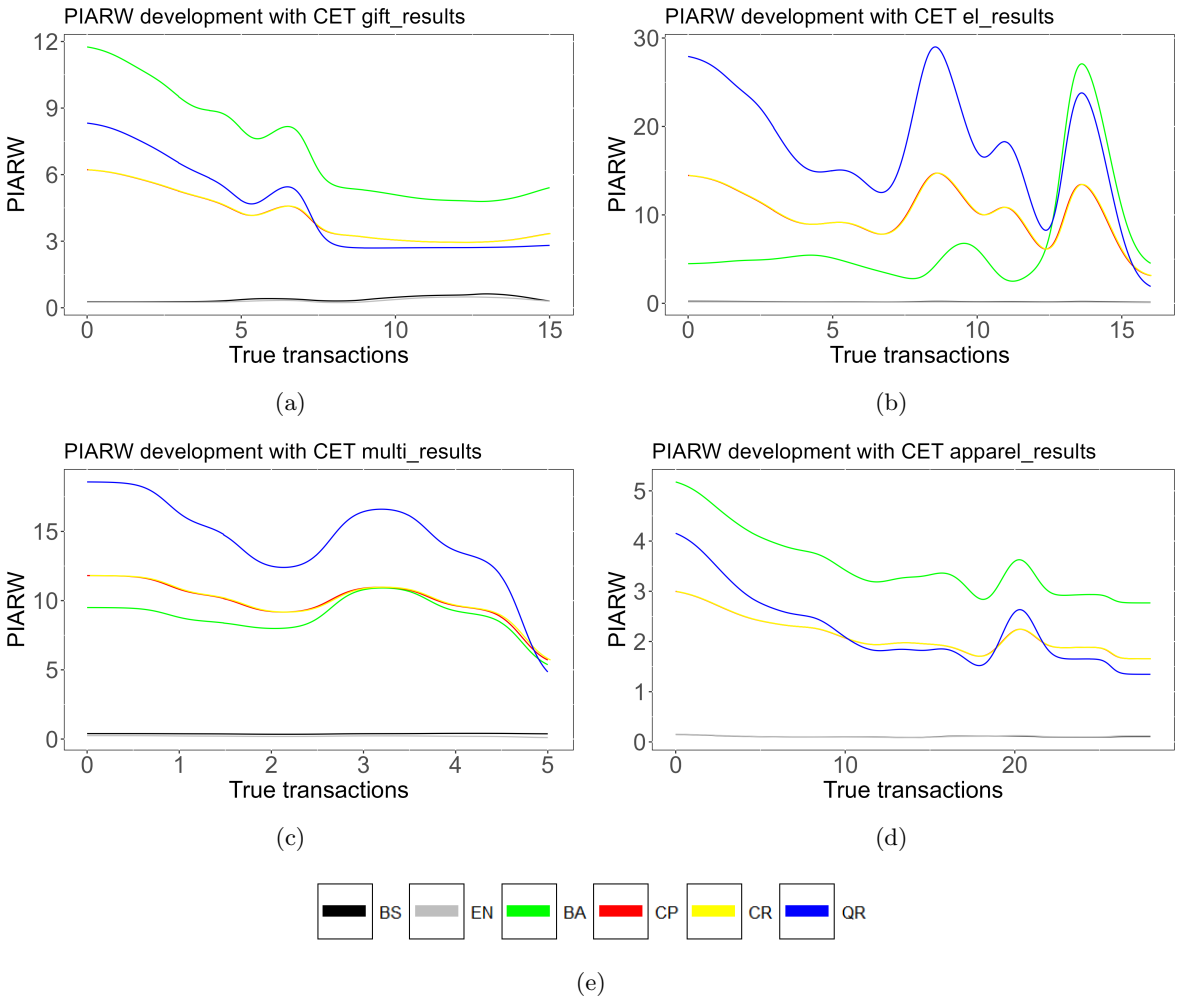
(a) (b) (c) (d) (e)

Figure 4: PICP development with CET results

Width

Considering PICP across CET levels, it must be seen in connection with width development. The situation across data sets is again very heterogeneous but also, there is generally a downward trend visible, while either BA or QR deliver the widest intervals and BS and EN deliver across all levels intervals with a length of nearly 0. The downward trend is not as dominant as for PICP but still visible. A lot of this phenomenon can be explained by scaling "normal"-sized intervals at lower levels with very small predictions, delivering relatively wider intervals, as indicated before in section 3.3.1. The key insight is therefore that a decreasing PICP with increasing customer value is first and foremost not the responsibility of relatively narrowing PIs.

The corresponding figures for the PIARW development for BS and EN only with adapted scale and the development of absolute width can be found in the appendix under 13a - 13d and 14a - 15d, respectively.



3.3.3 Performance over varying training and prediction periods

Motivation

The analysis so far has held the learning and prediction period for each data set constant and examined how performance measures vary across methods and CET levels. This section will deal with the case if periods for learning and prediction are varied and will assess how only the PICP is influenced. This examination is mainly motivated by the methods QR, CP and CR which use data from a previous cohort. These data exist for the data sets used in this work, but in reality, a lack of past data is not unrealistic and might cause problems. I.e. the data could be biased (e.g. from a too short time period) so that inappropriate quantiles (CR and CP) are learnt, or the wrong parameters are selected (QR). As this topic is not the core of this work, it will be held concise and concentrate on the most central insights.

Implementation

Due to computational reasons, the analysis will be limited to the electronics and gift data sets and the bootstrap based methods will be left out because only coverage is assessed here and a loss in this area is more relevant to BA, QR, CP and CR. For both data sets, around 50 combinations of learning and prediction times for the old and the new cohorts are used.[11] For each period combination, a model is fitted, predictions are made, and prediction intervals are derived and assessed regarding coverage on the current cohort. The following graph resulted from this procedure.

[11] As BA does not use data from an old cohort, only learning and holdout period of the current cohort were changed here.

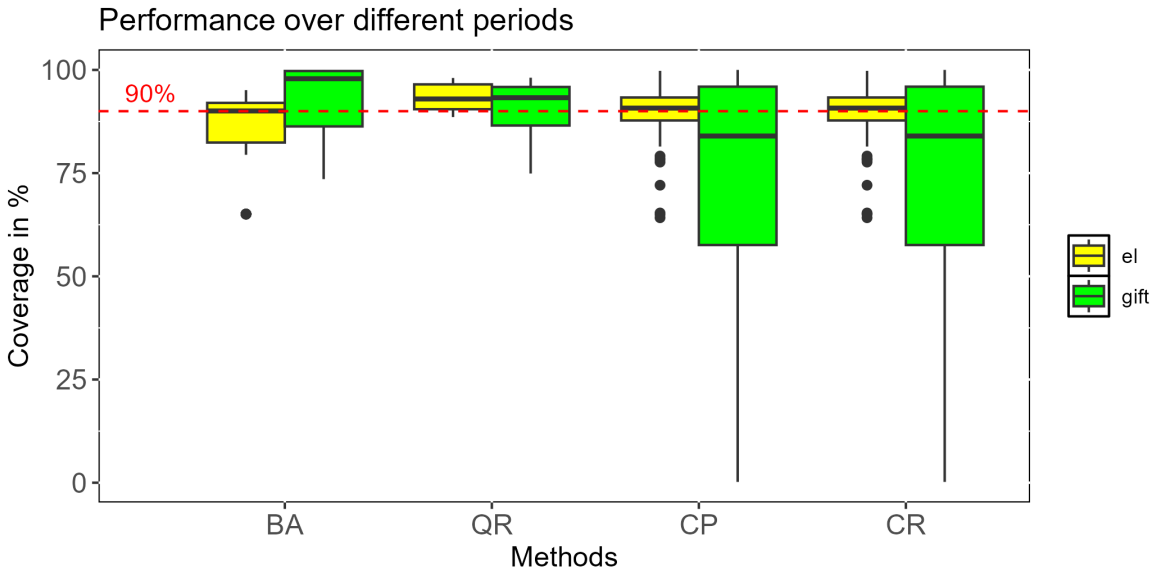


Figure 6: Method performance over different periods

BA and QR are, regarding coverage, not affected by changing period lengths or data sets and deliver constantly around 90% coverage. In contrast, the performance of the conformal prediction implementations suffer for the gift data set a lot while delivering decent results for the electronics data set. Looking directly into the data (see these links for the gift and el data set, otherwise available on request) gives several insights but no definite explanation for the underperformance and especially not for the difference between the two data sets. Typically, low coverage resulted from short learning or prediction periods for the old cohort, what makes intuitively sense as it is here where the quantiles are derived. On the other hand, one can also observe cases in which all periods, for both cohorts have been low which performed decently. It suggests that, when both cohorts have been treated equally in terms of model fitting and quantile forming conditions, the method works properly.
The motivation for this chapter was to give an intuition how stable the methods are and if managers can apply them safely. BA and QR seem to work stable, regardless of data set or learning and prediction time. The conformal prediction-based methods seem to suffer on one dataset but work well on the other one. The reasoning behind it cannot be clarified totally but tendencies are observable. It will be necessary to use more runs and data sets to get deeper insights for all methods but one should be cautious when using CP or CR in combination with varying learning and prediction periods.

3.3.4 Results summary

See below table 10 which summarizes and evaluates different aspects of the methods' performance and their observed capabilities from this section.

Method	Adequate coverage	Consistent coverage across CET levels	Consistent width (relative) across CET levels	Consistent performance across data sets	PICP independent from learning and prediction periods	Usefulness	Computational intensity
BS	No	Yes (but low)	Yes	Yes	Not tested	Low	Medium
EN	No	Yes (but low)	Yes	Yes	Not tested	Low	Low
BA	Yes	No	No	Yes	Yes	Medium	High
QR	Yes	No	No	PICP yes, width no	Yes	Medium	High
CP	Yes	No	No	Yes	No	Medium	Low
CR	Yes	No	No	Yes	No	Medium	Medium

Table 10: Results summary table

3.4 Identification of top customers

As a final step, it will be discussed whether and how intervals can be used beyond the assessment of model uncertainty. Can PIs help identify valuable customers? To answer this question, different customer identification metrics which incorporate the additional information from previously derived intervals are used. See the following approach

3.4.1 Basic approach

Implementation

Step 1: Define appropriate metrics to rank customers, incorporating new information coming from intervals

Metric 1: **Benchmark: hpp**: Highest Point Predictor (CET) coming directly from the PNBD model

Metric 2: hub: Highest Upper Boundary

Metric 3: hiw: Highest Interval Width

Metric 4: huu: Highest Upwards Uncertainty (the difference between the CET and the upper boundary)

Metric 5: htp: Highest Three-Point Estimate: The lower boundary, the CET and the upper boundary of a customer i are weighted equally $\left(\frac{1}{3} * (LB_i + CET_i + UB_i)\right)$

Metric 6: csw: The CET is squared and divided by the interval width $\left(\frac{CET_i^2}{UB_i - LBi}\right)$

Metric 7: ssq: The CET is squared and divided by the square root of the interval width $\left(\frac{CET_i^2}{\sqrt{UB_i - LB_i}}\right)$

Step 2: Apply these metrics to all customers across all methods and data sets

Step 3: Rank customers for each metric, data set and method

Step 4: For each data set and method, pick the top x%[12] of the customers according to the respective metric

Step 5: Compare the selected customers with the actual top performing customers

The following table shows the results for each data set, method and metric.
Note: max_rel states how much % of the customers are considered "Top-customers", which should be ideally close to 10%. max_abs is the actual number of "Top-customers" in the respective dataset. All values for the previously introduced metrics tell how much percent of respective "Top-customers" have been identified by this metric. Example: For the gift data set, 6% of all customers were identified as "Top-customers" (in approximation to the desired 10%) and relying on hpp, one would identify 22.61% of those 6%.

Data	Method	max_rel	max_abs	**hpp**	hub	hiw	huu	htp	csw	ssq
gift	BS	0.06	127	0.2261	0.2422	0.2422	0.2422	0.2342	0.1373	0.218
	EN	0.06	127	0.2261	0.2261	0.2342	0.2422	0.2261	0.2422	0.2342
	BA	0.06	127	0.2261	0.2342	0.2342	0.2019	0.218	0.1938	0.218
	QR	0.06	127	0.2261	0.2584	0.2584	0.1857	0.2422	0.2342	0.2342
	CP	0.06	127	0.2261	0.2261	0.2261	0.2261	0.2261	0.2261	0.2261
el	BS	0.22	1088	0.2414	0.2432	0.2170	0.2170	0.2414	0.2348	0.2395
	EN	0.22	1088	0.2414	0.2414	0.2133	0.2348	0.2404	0.2376	0.2395
	BA	0.22	1088	0.2414	0.2535	0.2535	0.2217	0.2320	0.3929	0.3929
	QR	0.22	1088	0.2414	0.2030	0.2030	0.1020	0.2273	0.2414	0.2423
	CP	0.22	1088	0.2414	0.2414	0.2414	0.2414	0.2414	0.2414	0.2414
multi	BS	0.08	313	0.2960	0.2960	0.3024	0.3024	0.2960	0.2960	0.2960
	EN	0.08	313	0.2960	0.2960	0.3057	0.2896	0.2960	0.2960	0.2960
	BA	0.08	313	0.2960	0.3314	0.3314	0.2574	0.2928	0.0676	0.0676
	QR	0.08	313	0.2960	0.2510	0.2510	0.2156	0.2928	0.2960	0.2960
	CP	0.08	313	0.2960	0.2960	0.2960	0.2960	0.2960	0.2960	0.2960
apparel	BS	0.12	332	0.3996	0.3996	0.3173	0.3115	0.3996	0.3497	0.3996
	EN	0.12	332	0.3996	0.3996	0.3056	0.3115	0.3996	0.3791	0.3996
	BA	0.12	332	0.3996	0.3732	0.3732	0.3644	0.3849	0.3967	0.3996
	QR	0.12	332	0.3996	0.3996	0.3996	0.0441	0.3996	0.4026	0.4026
	CP	0.12	332	0.3996	0.3996	0.3996	0.3996	0.3996	0.3996	0.3996

Table 11: Benchmarking the highest point predictor hpp against other metrics in its capability to identify especially valuable customers

For a better overview, the results are summarized in table 12:

Metric (compared to hpp)	hub	hiw	huu	htp	csw	ssq
Better or equal	0.85	0.65	0.35	0.65	0.6	0.75
Better	0.30	0.40	0.15	0.10	0.20	0.25
Worse	0.15	0.35	0.65	0.35	0.4	0.25
Mean advantage (rel)	0.0025	-0.0263	-0.1372	-0.0044	-0.0396	-0.0074
Sd of advantages	0.0191	0.034	0.0837	0.0068	0.0665	0.0629

Table 12: Summary of table 11

[12]The target in this work is to pick the 10% customers with the highest true value. That is not always possible because the number of purchases has only few discrete outcomes, so that e.g. the last customer from the top 10% has 4 purchases and the next one, outside these 10%, also has 4 purchases. It would not be reasonable to include one and exclude the other. So, 10% cannot always be perfectly achieved and the respectively closest realizable number is taken as approximation and reported in the overview as max_abs.

The results are throughout all metrics not satisfying. The best metrics usually identify between 20% and 40% of the top-customers. Those are in most of the cases the benchmark hpp, the highest upper boundary hub, the highest interval width hiw and the three-point estimate htp. The latter three sometimes gain single digit percent over hpp but also lose them on another metric-data set combination so that there is no exploitable pattern. The results from csw and ssq vary more and beat the benchmark in 1 case significantly but also lose significantly in other ones without any recognizable pattern. huu is the only metric that loses on average more than 10% compared to hpp. It is noteworthy that all metrics usually perform similarly good (bad) across methods what was not to expect, regarding the difference in coverage power of the methods.

Explanation
The PI generating methods incorporate the uncertainty from the whole data set and then apply it to single predictions to construct intervals. This holds for all methods, regardless of their performance in terms of coverage or width. Especially,

- **BS**: The Bootstrap method is focused on capturing parameter uncertainty which is not customer-specific. With each bootstrap sample, a new model with new parameters is derived but it will be driven by the entirety of customers in the sample and will not hold any individual information about a customer. The resulting quantiles are based on these bootstrap predictions and do therefore not hold customer-specific information.
- **EN**: In the Ensemble method, first there is 1 "root" model fitted which has exactly 1 covariance matrix that represents the amount of uncertainty within the parameter estimates. This uncertainty comes from all customers and is therefore not unique to any single customer and neither are the models that imitate this uncertainty. The resulting intervals will not contain any individual uncertainty.
- **BA**: The intervals using BA come from the posterior predictive distribution of outcomes which is derived from the posterior distribution of model parameters. The latter is for sure based on the data that came from the customers, but it does not hold any information about individual customers. However, this comes at a surprise, as the BA intervals were the only ones to show variation among customers with similar CET values.
- **QR**: For QR, the whole information about uncertainty of customers is summarized by eigth parameters, four for the lower and four for the upper boundary. It is not possible to retrieve information for individual customers from this point.
- **CP/CR**: Here, all the information about uncertainty from an old cohort is summarized by a single quantile which is then applied to new customers and scaled by their individual CET. This procedure does by definition not include any information about individual uncertainty, except the level of a point prediction, which is not helpful at all when the goal is to differentiate between customers with similar point predictions.

3.4.2 Approach using a covariate

Following the argumentation in the previous section, then intervals do not help to identify especially valuable customers because they do not hold information about uncertainty associated with individual customers. They rather represent the uncertainty coming from a whole data set and its predictions and apply it to individual customers.
Hence, this problem could be resolved by introducing a covariate, that informs about the model uncertainty for individual customers, and incorporating it into PI construction. Individual uncertainty can be understood as difference between CET (model prediction) and the true number of repurchases, which PIs attempt to cover. Another, more intuitive, wording for this individual customer uncertainty is therefore *the deviation between customers' expected behavior and their actual behavior*. To continue this idea, it is essential to have a covariate that represents, at least partially, this deviation. Unfortunately, none of the used data sets includes such a covariate and it needs to be simulated what makes this section rather a proof of concept than an actual application. Regarding the concrete implementation, see the following procedure that is applied on all methods and data sets.

Step 1: Simulate the covariate

- Step 1.1: Calculate the absolute difference between the CET and actual number of repurchases (for each customer of interest)
- Step 1.2: Pick a level of correlation between this absolute prediction error and the prospective covariate[13]
- Step 1.3: Simulate a numeric[14] covariate that is correlated with the absolute error at the level set in Step 1.2

Step 2: Reconstruct the PIs, incorporating the new covariate

- Step 2.1: Calculate the median of the covariate
- Step 2.2: Create a factor by dividing every customer's individual covariate value by the median from 2.1, resulting in a smaller factor for customers with lower deviation and a higher factor for customers with higher deviation
- Step 2.3: Multiply each customer's initial interval width with the individual factor to get the new interval length that incorporates the individual uncertainty
- Step 2.4: Divide it by 2 and add/subtract it to the point prediction what gives the new individual interval

Step 3: With the newly created intervals, conduct the steps from the Basic approach in 3.4.1

This procedure is very straight-forward, easy to implement for different methods and offers several options for adaptations within the process, e.g simulating the covariate differently, conducting the scaling differently or including tolerances[15]. Also, there might be ways to directly include a covariate into the original PI fitting procedures what could be subject to future research.

General PI performance

Besides the application in identifying especially valuable customers, it is interesting to see how the additional uncertainty information impacts the overall performance of PIs, compared to the regular version.

The performance looks very similar to the regular case in terms of PICP and PIARW, see figure 7. EN and BS keep their clear undercoverage while also the other methods produce nearly as reliable intervals as before. However, the picture is a bit more scattered for the four reliable methods, in particular for BA where the electronics data set falls considerably below 90% coverage, due to the reason raised in the previous section, see footnote 15. Also CP and CR have a tendency to slightly undercover. A second insight concerns the width. Especially the four methods with a high reliability used to struggle with high interval width before the introduction of this covariate. All of them produce visibly shorter intervals now, what makes intuitively sense. It shows a distinct advantage of incorporating knowledge about individual uncertainty, as shorter intervals are preferred over wider intervals with approximately constant coverage.

[13] The higher, the better the results are supposed be, but also the harder it will be in practice to find such a covariate. In this work, $r = 0.6$ was used.

[14] In theory, also other types like categorical should work but they would require a different approach.

[15] A potentially useful tolerance could be to set the new lower boundary to 0 if it close to 0. Many customers will have 0 repurchases and if the model predicts something close to 0, the interval will be scaled down, causing the 0 to fall below the lower boundary what distorts method performance in terms of PICP. That should not be an issue in a real world application when the covariate is not simulated.

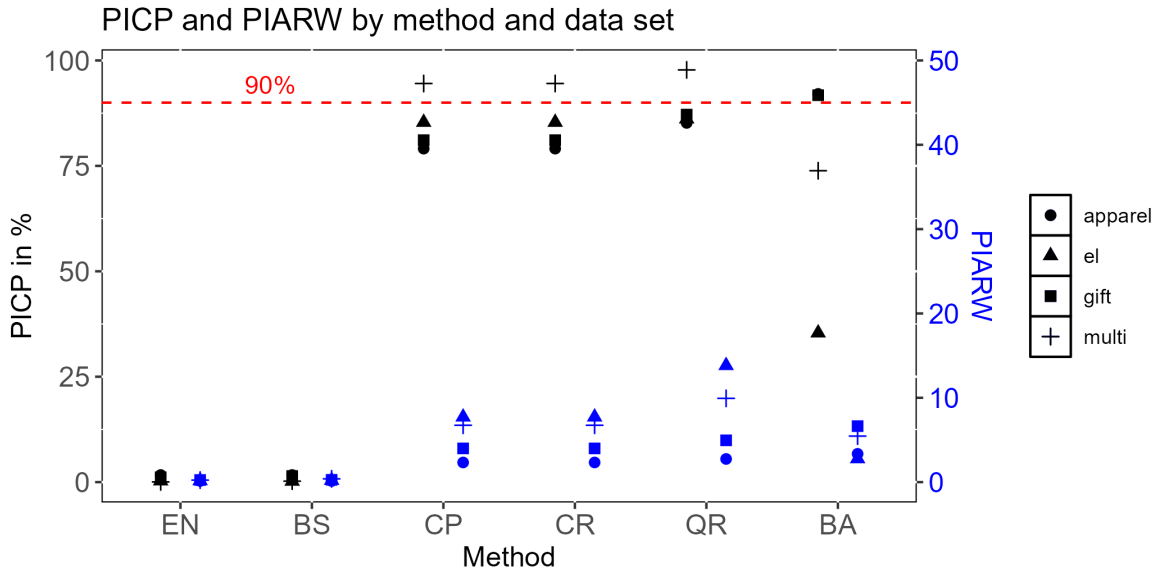


Figure 7: PICP and PIARW by method and data set (with covariate)

The motivation behind this section is to see how the introduction of individual uncertainty information changes the previously calculated PIs and if they can be used consecutively to identify especially valuable customers. In figure 8, the first part can be observed. The new, individual intervals have varying lengths for the same method and customers of similar CET, what is pre-requisite, when differentiating customers beyond their point predictor.

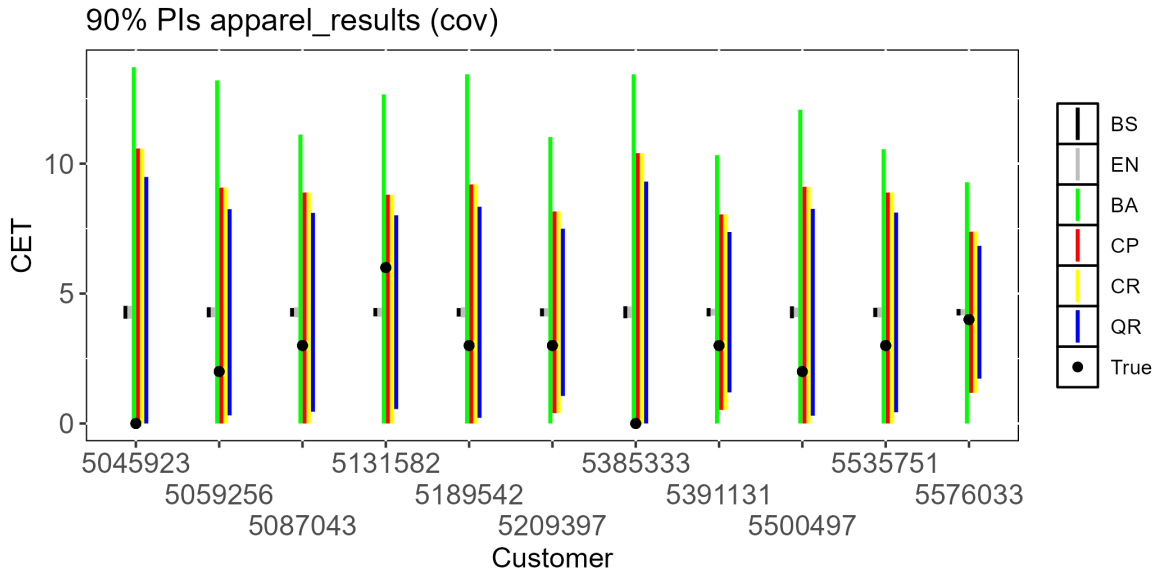


Figure 8: 90% PIs apparel results (with covariate)

Customer selection
In this final section, it will be evaluated if the newly individualized intervals can indeed help identify particularly valuable customers. The exact same approach as in section 3.4.1 is used to ensure comparability.

Data	Method	max_rel	max_abs	**hpp**	hub	hiw	huu	htp	csw	ssq
gift	BS	0.06	127	0.2261	0.2584	0.2907	0.2907	0.2261	0.0161	0.1130
	EN	0.06	127	0.2261	0.2503	0.3149	0.3149	0.2261	0.0404	0.1453
	BA	0.06	127	0.2261	0.3391	0.3391	0.3230	0.3068	0.0807	0.1696
	QR	0.06	127	0.2261	0.3472	0.3472	0.3795	0.3230	0.1130	0.2019
	CP	0.06	127	0.2261	0.3230	0.3230	0.3068	0.3149	0.1130	0.2019
el	BS	0.22	1088	0.2414	0.2414	0.2124	0.2124	0.2414	0.2376	0.2423
	EN	0.22	1088	0.2414	0.2404	0.2021	0.2021	0.2414	0.2423	0.2442
	BA	0.22	1088	0.2414	0.2152	0.2152	0.2058	0.2217	0.3929	0.3929
	QR	0.22	1088	0.2414	0.2283	0.2142	0.2348	0.2320	0.2414	0.2432
	CP	0.22	1088	0.2414	0.2236	0.2236	0.2198	0.2273	0.2442	0.2423
multi	BS	0.08	313	0.2960	0.2960	0.3121	0.3121	0.2960	0.2767	0.2928
	EN	0.08	313	0.2960	0.2960	0.4472	0.4472	0.2960	0.2864	0.2928
	BA	0.08	313	0.2906	0.3282	0.3282	0.3153	0.3185	0.0676	0.0676
	QR	0.08	313	0.2960	0.3764	0.3764	0.4376	0.3571	0.2928	0.2928
	CP	0.08	313	0.2960	0.3314	0.3314	0.3604	0.3185	0.2928	0.2928
apparel	BS	0.12	332	0.3996	0.3996	0.3144	0.3144	0.3996	0.3144	0.3673
	EN	0.12	332	0.3996	0.3996	0.3232	0.3232	0.3996	0.3262	0.3761
	BA	0.12	332	0.3996	0.3879	0.3791	0.3996	0.3996	0.3673	0.3908
	QR	0.12	332	0.3996	0.4114	0.3732	0.3849	0.4055	0.3702	0.3937
	CP	0.12	332	0.3996	0.4114	0.3849	0.3791	0.4026	0.3644	0.3908

Table 13: Benchmarking the highest point predictor hpp against other metrics in its capability to identify especially valuable customers (with individualized PIs)

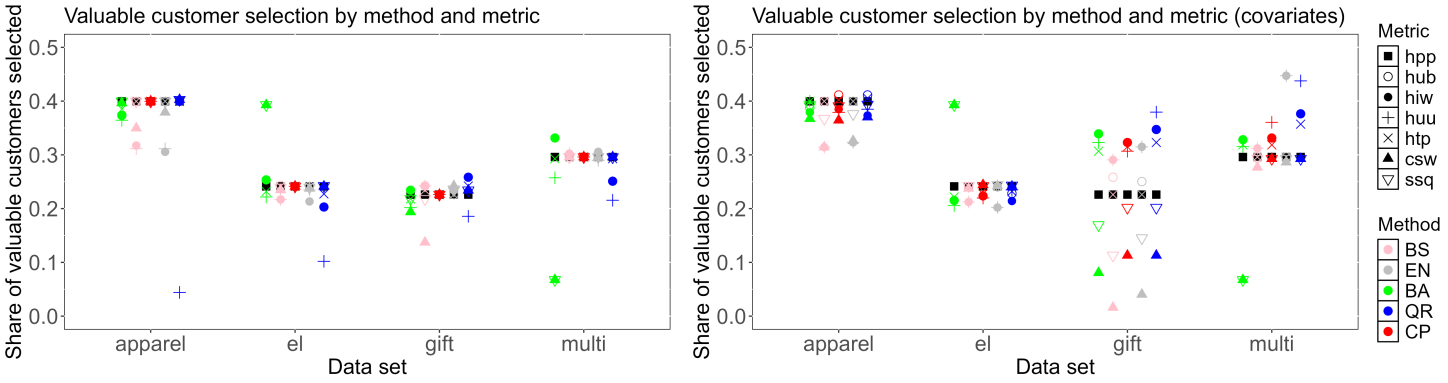
In this new approach, it is by far more likely for several metrics and methods to beat the benchmark hpp. While hub, hiw, huu and htp perform nearly always better or equal than the benchmark, except the electronics data set, it is the opposite case for csw and ssq which only perform well, across methods in the electronics data set. It is hard to tell why exactly this is but in figure 1 and figure 7, it has always been this data set which had the longest intervals but not necessarily a higher coverage. The findings in this section confirm that the PIs and their further application seem to struggle with this particular data set. See the results summarized in table 14.

Metric	hub	hiw	huu	htp	csw	ssq
Better or equal	0.75	0.50	0.55	0.85	0.20	0.25
Better	0.50	0.50	0.50	0.40	0.15	0.25
Worse	0.25	0.50	0.45	0.15	0.80	0.75
Mean advantage (rel)	0.0245	0.0219	0.0274	0.0169	-0.0568	-0.0231
Sd of advantages	0.044	0.0687	0.0733	0.0352	0.0895	0.0688

Table 14: Summary of table 13

Tables 11 and 13 are visualized in figure 9. Without using a covariate (left), only very few exemptions beat the benchmark and that rather by chance than with a recognizable pattern. The likelihood of selecting an adverse combination of method and metric and falling below the benchmark appears higher than picking a favorable combination. The reason is that all metrics only include information from the upper and lower boundary and the hpp itself, where the hpp is the only ingredient that holds individual information. Therefore, only few deviations from the benchmark are visible. For the apparel, electronics and multi data set, the benchmark is barely beaten at all. When using individualized quantiles (right), the picture is slightly better for the apparel and electronics data set but still, it is difficult to beat the pure point predictor, here. In contrast, for the gift and multichannel data set, one can

observe a significant spread of achieved values and therefore, chances to beat the benchmark. While csw and ssq are loosing visibly on the gift data set across methods, there are numerous opportunities to be better than hpp with other metrics. For the multichannel data set, nearly all combinations of metrics and data sets are comparable with hpp or distinct better. Without further investigations, it is hard to tell why this approach works for some data sets but not for all.



(a) Without covariate

(b) With covariate

Figure 9: Identifying valuable customers without and with individualized PIs across methods and metrics

What is a good strategy for marketing managers in a real-world scenario?

- Without covariate
  - Figure 9a) and table 11 apply.
  - Chances are low to pick a metric and method that beat the benchmark significantly. Downside risks exist and are more likely to be realized than upside potential.
  - Without further information, it is reasonable to stay with the point predictor or decide for hub (with anything but QR), hiw (with QR or CP), huu (with CP) or htp (with BS, EN or CP) while none of these strategies is expected to bring a clear advantage over hpp.
  - Note that this analysis was limited to four data sets. The performance of the suggested strategies may vary on other data sets.
- With covariate
  - Figure 9b) and table 13 apply.
  - It can be beneficial to use alternatives to hpp but which to choose strongly depends on risk aversion, data set and possible experience with past data.
  - Conservative: hub usually outperforms hpp on 3 out of 4 examined data sets, so that gains are more substantial than the losses from the remaining data set. htp has a similar behavior to hub but is in most cases equal or inferior.
  - Risk-taking: hiw and huu offer with BA, QR and CP a similar overall performance as hub but have a greater variance, hence more upside potential but also downside risk.
  - Special cases: When one knows exactly which methods work well for the own business, it can make sense to use special combinations like csw and ssq, especially with BA as that is the only way to beat the benchmark for the electronics data set significantly. Similarly, hiw and huu with EN (and QR) can beat the benchmark on the multichannel data set more distinct than others.

Sidenote: For this implementation, a correlation of $r = 0.6$ was chosen, which is admittedly a high value that could be difficult to find in a real-world data set. However, it shows the power of such a potential covariate and that it could indeed enhance the results significantly. For completeness, an overview is provided how different values for r would influence the result, see figure 16 in the appendix.

4 Discussion and conclusion

The main objective of this work was to assess the uncertainty which is associated with the PNBD model's predictions of the individual customer lifetime value. Previous research on this topic was found to be scarce. To fill this gap, several methods that produce prediction intervals were identified in the literature, adapted and applied to the specific context of CLV prediction. Amongst others, a special focus was put on Conformal Prediction as a relatively young method that has recently gained attention in the statistical community. The resulting intervals were benchmarked with several metrics against an existing PI generating bootstrap method from the R package CLVTools [4].
A central contribution of this thesis is the successful implementation of methods that deliver intervals with high reliability on several real-world data sets, capturing the underlying uncertainty, which no other research has covered so far. It were especially the Bayesian Approach, Quantile Regression and Conformal Prediction which achieved a reliable coverage, which, however, comes at a cost. Capturing uncertainty adequately in the presence of high uncertainty causes wide intervals and little sharpness, which is the case for the PNBD model. Two other methods, including the bootstrap implementation from CLVTools, in contrast, create very short intervals, leading to low reliability but high sharpness. No method was found that provides a compromise between these two criteria.
Another objective was to assess the capacity of the constructed intervals to improve the distinction between low- and high-value customers with similar CET. Unfortunately, the PI generating methods use mostly information about the uncertainty of the whole data set and then apply it to individual customers to form PIs. This lack of individual uncertainty information can be overcome by introducing a covariate that holds information about individual uncertainties. In this work, this covariate needed to be simulated but might exist for other data sets. With this additional information, the intervals were re-scaled and individualized and could eventually be used to enhance the differentiation of customers. In the course of this work, there was no method found that achieves both, high reliability and sharpness at the same time and none of the used methods incorporates directly individual uncertainty. In addition, Quantile Regression and Conformal Prediction need past data to work, what might pose a major obstacle in real-world scenarios. Regarding the application in identifying especially valuable customers, it was necessary to simulate a covariate, as the data sets used did not provide a suitable one. Whether such a covariate actually exists in practice is not certain but identifying one and verifying the results from this work in reality could be subject to future research. Also, the improvement of PI generating methods, in terms of e.g. incorporating directly individual uncertainty, improving speed, especially for the Bayesian Approach, or developing a method that offers a good compromise between reliability and sharpness are areas that need further attention. This work utilized four data sets to ensure the generalizability of the results. However, confirming the findings with additional datasets from different industries and exhibiting diverse customer behavior would be beneficial.

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5 Appendix

Method	PICP	ACE	PICPW	PIARW	PIARWW	MSIS	SWR	Upper cov.	Lower cov.	Time [s] (abs)
BS	0.8606	0.0099	0.5162	0.4207	0.4815	0.6918	1.1275	0.1501	0.5536	0.8262
EN	1.0262	0.0099	0.7570	0.2810	0.3658	0.6917	1.2363	0.1488	0.5557	0.4662
BA	0.0483	-0.8839	0.2500	0.4518	0.3901	1.1326	0.4069	0.0485	0.0021	1.6797
QR	0.0341	-0.7262	0.1232	0.7283	0.6986	0.7921	0.8409	0.0336	0.0005	0.0995
CP	0.055	-1.8659	0.1911	0.5888	0.5552	0.9015	0.7315	0.0504	0.0073	0.8851
CR	0.055	-1.8659	0.1911	0.5888	0.5552	0.9015	0.7315	0.0504	0.0073	0.7446

Table 15: Variation table (coefficient of variation)

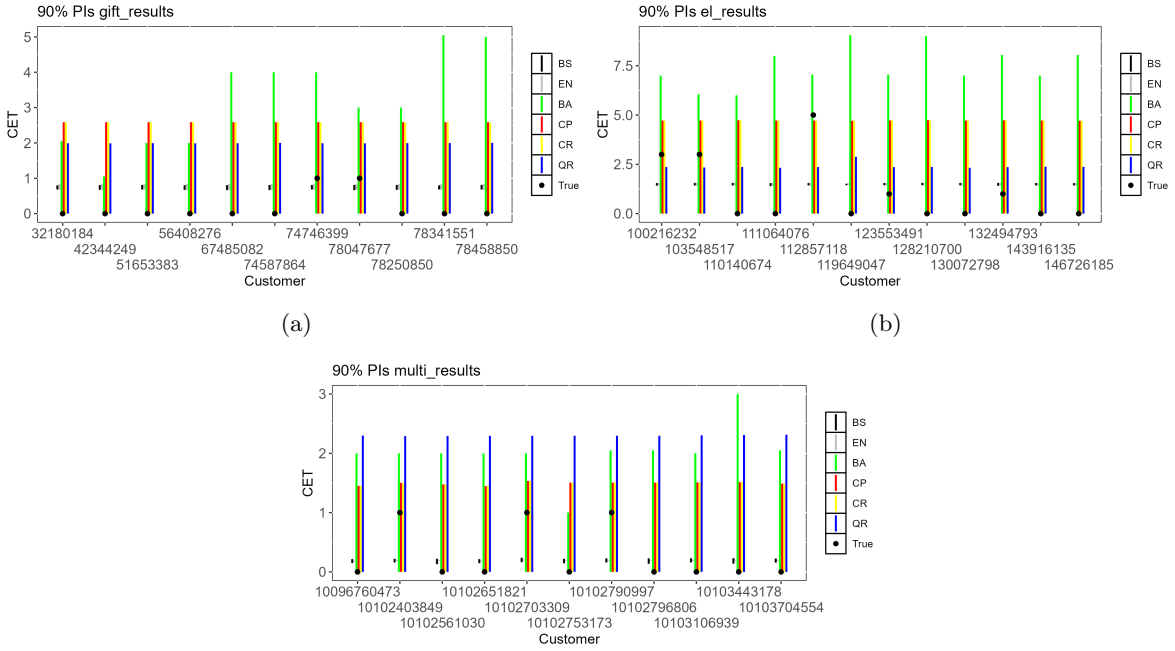
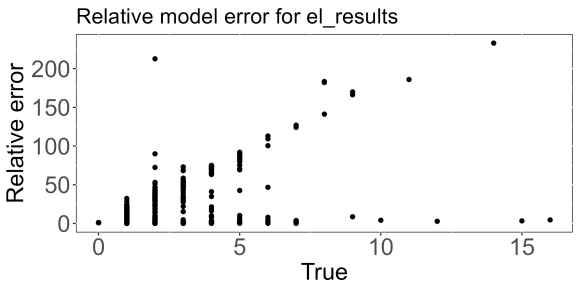


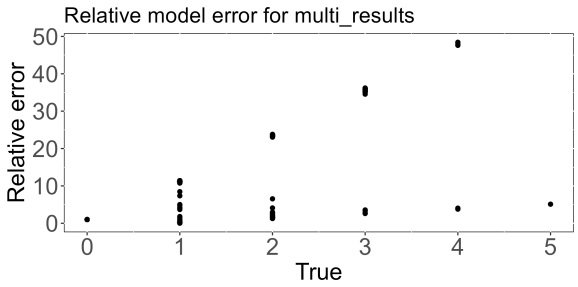
Figure 10: 90% PIs for selected customers for gift, el and multi



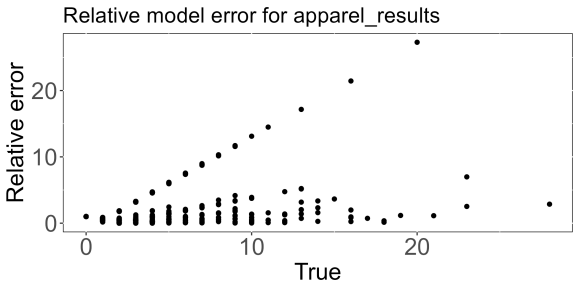
(a) gift data set



(b) electronics data set

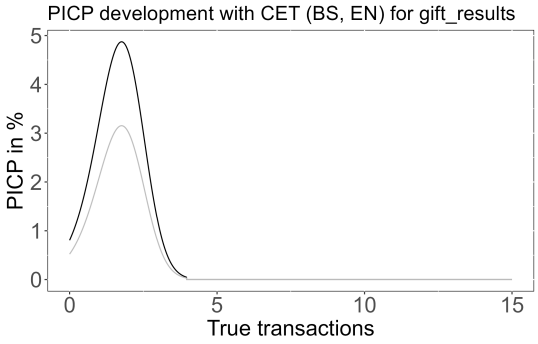


(c) multichannel data set

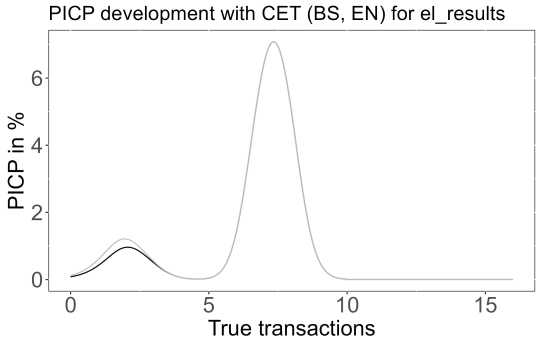


(d) apparel data set

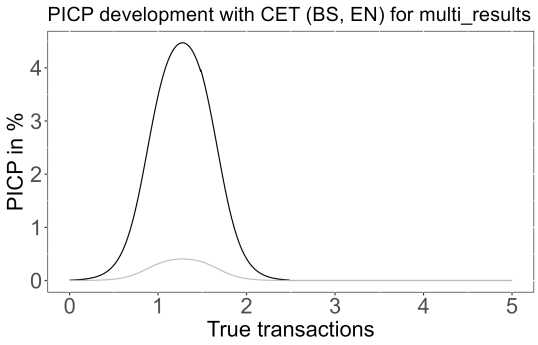
Figure 11: Relative model error



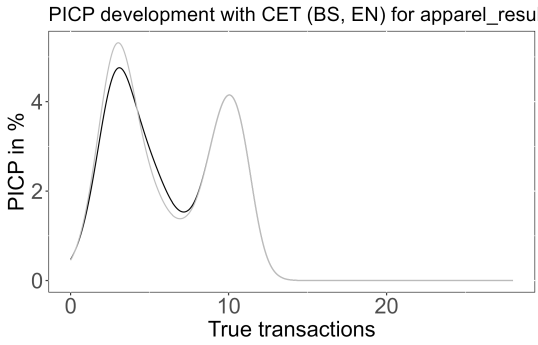
(a) gift data set



(b) electronics data set

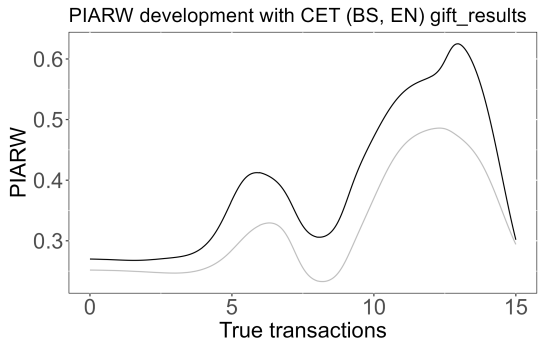


(c) multichannel data set

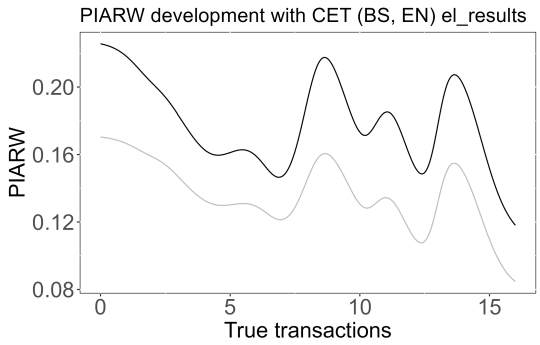


(d) apparel data set

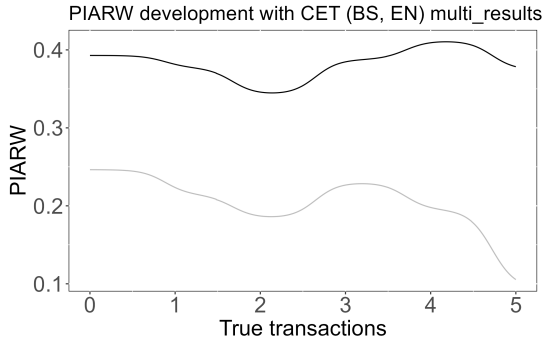
Figure 12: PICP development with CET for BS and EN only



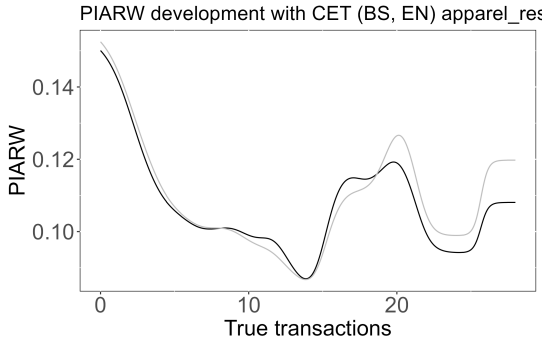
(a) gift data set



(b) electronics data set

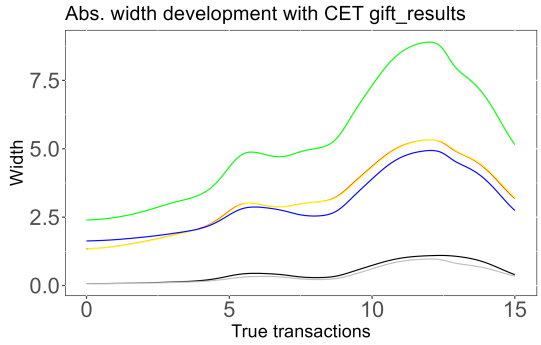


(c) multichannel data set

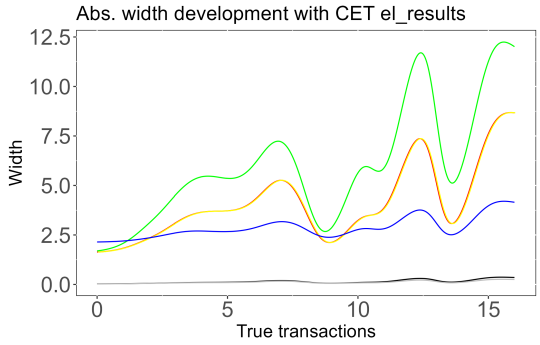


(d) apparel data set

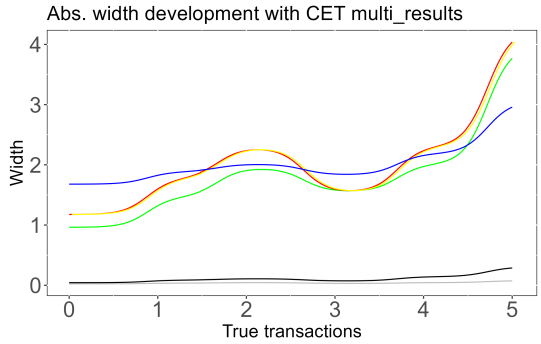
Figure 13: PIARW development with CET for BS and EN only



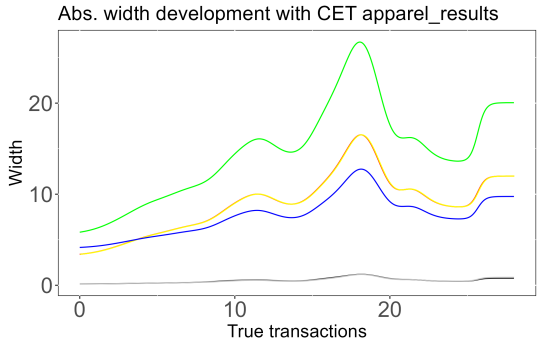
(a) gift data set



(b) electronics data set

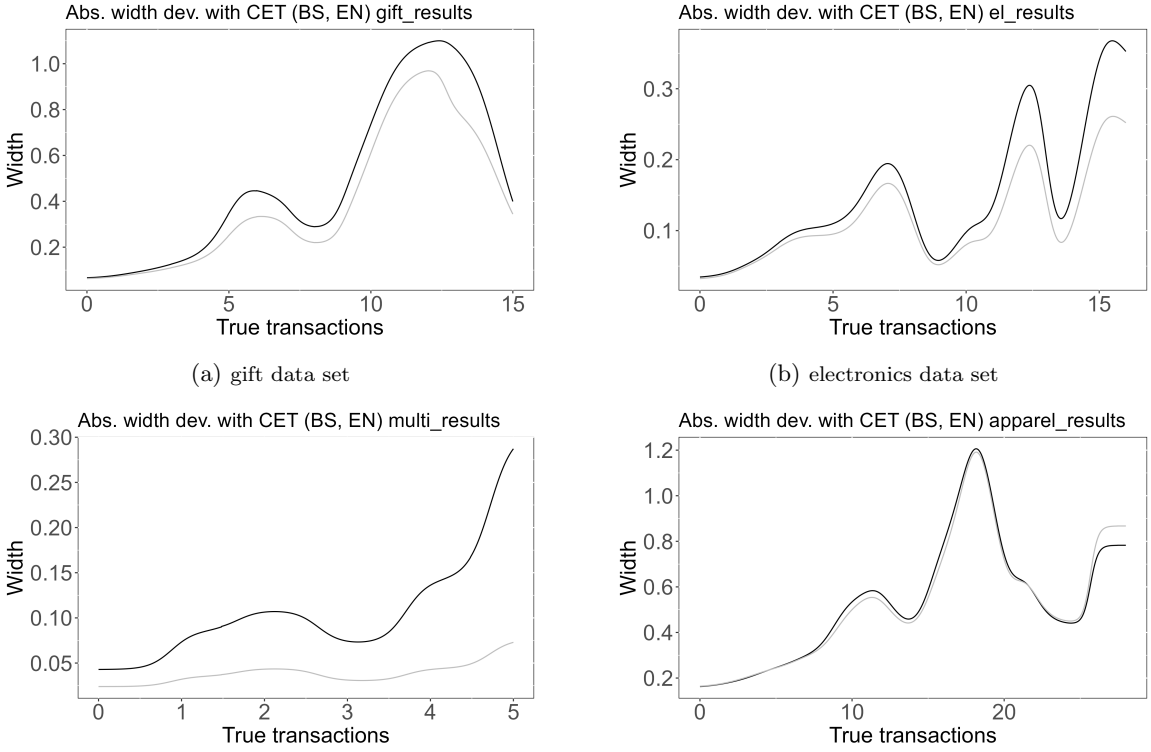


(c) multichannel data set



(d) apparel data set

Figure 14: Absolute width development with CET



(a) gift data set

(b) electronics data set

(c) multichannel data set

(d) apparel data set

Figure 15: Absolute width development with CET for BS and EN only

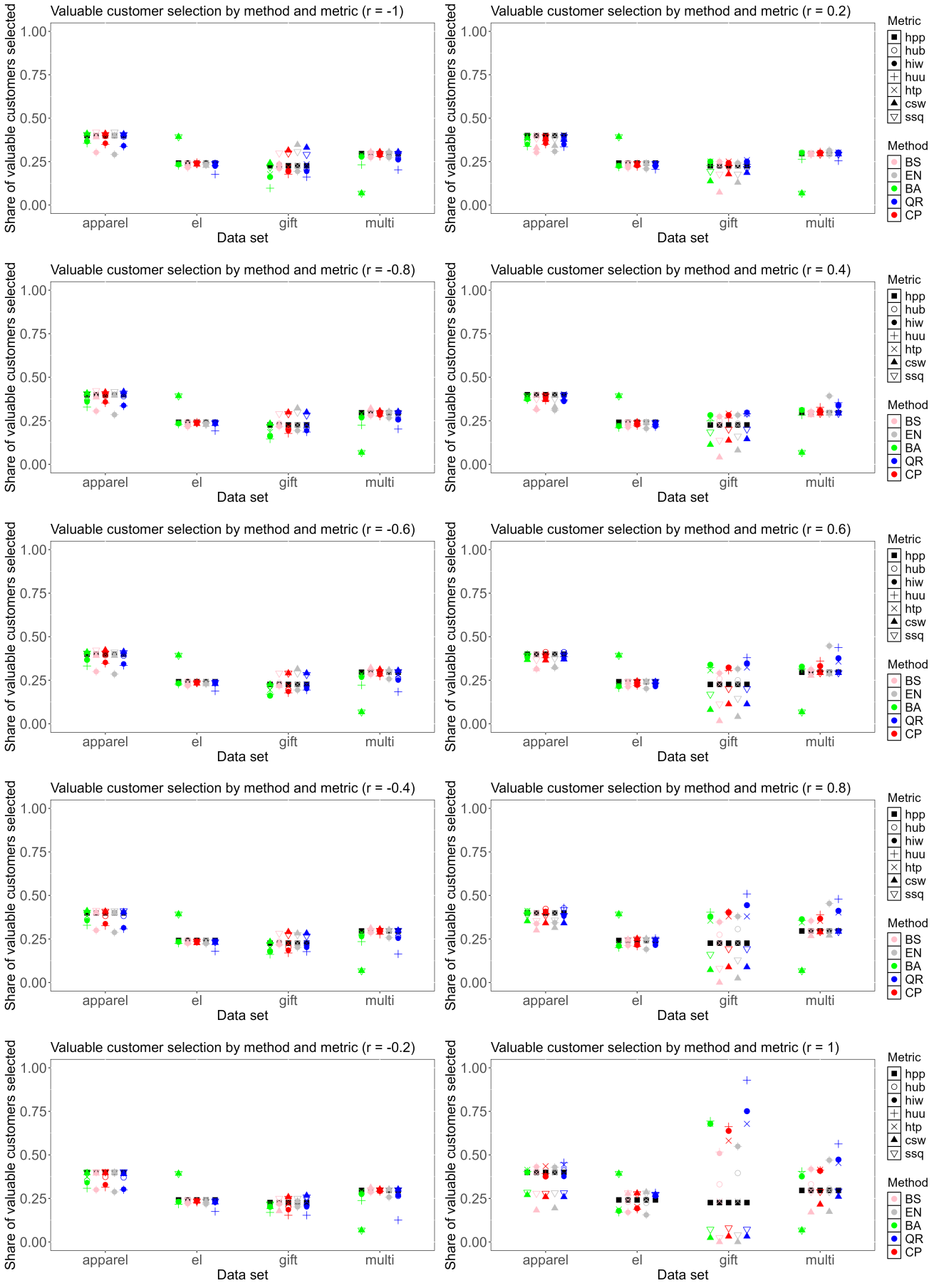


Figure 16: Identification capabilities for changing correlation with the covariate